

# TITLE OF YOUR PROJECT

## A PROJECT REPORT

Submitted in partial fulfillment for the Degree of  
Bachelor of Technology under the  
School of Computing

Submitted by

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# Acknowledgements

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# Abstract

Write your abstract here

# Chapter 1

## Introduction

### 1.1 Background

The global aging population is experiencing a rapid increase in the prevalence of chronic diseases, placing an unprecedented burden on healthcare systems worldwide. Conditions such as hypertension, diabetes, depression, obstructive sleep apnea, and respiratory disorders are disproportionately common among elderly individuals and are closely intertwined with disruptions in sleep physiology. Sleep, far from being a passive biological state, is a rich source of physiological information that simultaneously reflects the functional status of the cardiovascular, respiratory, and neurological systems.

Polysomnography (PSG) is the clinical gold standard for capturing this multi-modal sleep physiology. A single overnight PSG recording yields continuous signals from multiple physiological channels including electroencephalography (EEG), electrooculography (EOG), electromyography (EMG), electrocardiography (ECG), respiratory effort channels (thoracic and abdominal), airflow, nasal pressure, and blood oxygen saturation ( $\text{SpO}_2$ ). Together, these signals encode temporal dynamics, spectral characteristics, and cross-modal coupling patterns that are clinically associated with a wide range of chronic pathological conditions.

Despite the diagnostic richness of PSG data, its routine clinical interpretation relies on labour-intensive manual scoring performed by trained sleep specialists, which is slow, expensive, and inaccessible in resource-limited healthcare settings. Automated deep learning approaches, while promising, are predominantly single-disease in scope, computationally intensive, and fail to generalise across heterogeneous elderly cohorts. Neural Architecture Search (NAS) and model compression techniques had remained largely unexplored for PSG-based health screening prior to this work, leaving the accuracy-efficiency trade-off unoptimised for edge deployment.

This project addresses these gaps through three progressively refined NAS strategies investigated on the Cleveland Family Study (CFS) dataset. The first strategy is Differentiable Architecture Search (DARTS), which applies gradient-based continuous relaxation over a directed acyclic graph cell structure to discover compact architectures under explicit FLOPs, MACs, and parameter budgets. The second strategy is Efficient Neural Architecture Search (ENAS), which replaces gradient-based search with a reinforcement learning controller trained via pol-

icy gradient to sample discrete architectures over a refined operation set, improving stability and robustness under severe class imbalance. The third strategy is a novel Hierarchical Neural Architecture Search (HNAS) framework driven by Proximal Policy Optimisation (PPO), which extends the search to a richer nine-channel multimodal setting with three-tier feature representation (temporal, spectral, and patient-level clinical), jointly exploring macro-level structural decisions and micro-level operation choices. The resulting system, named *SleepRisk AI*, additionally incorporates a real-time inference application with an AI-assisted chatbot (Sleep Copilot) and automated medical report generation, enabling clinically interpretable health risk assessment without requiring technical expertise from end users.

## 1.2 Problem Addressed

Manual PSG analysis suffers from well-documented limitations: it is time-consuming, highly dependent on specialist availability, prone to inter-rater variability, and economically prohibitive for large-scale screening. In resource-constrained healthcare environments — particularly those serving elderly rural or semi-urban populations — access to trained sleep specialists is severely limited, making routine PSG analysis practically infeasible.

Existing automated AI models for PSG analysis are predominantly designed for single-disease classification, typically targeting obstructive sleep apnea or sleep staging in isolation, and rely on large computationally expensive architectures that demand high-end GPU infrastructure for inference. Neural Architecture Search and model compression techniques had remained largely unexplored for PSG-based health screening, leaving the accuracy-efficiency trade-off entirely unoptimised for edge deployment.

The first NAS strategy investigated, DARTS, demonstrated that gradient-efficient architecture search could discover a lightweight cell-based model (0.17M parameters, 167.8M FLOPs) achieving strong multi-label AUC-ROC across cardiovascular, mental health, and respiratory disease cohorts on the CFS dataset using seven physiological channels. However, DARTS exposed two core limitations specific to the PSG multi-label setting. First, the gradient-based bi-level optimisation objective maximised overall accuracy, producing architectures biased towards the majority class and exhibiting near-zero recall for low-prevalence conditions such as depression and asthma. Second, DARTS training exhibited sensitivity to architectural bias and operational instability, with a strong tendency to favour skip connections that degraded the effective representational depth of discovered architectures.

ENAS was subsequently investigated to address these limitations through discrete, reinforcement learning-based architecture exploration. Using a recurrent controller trained via policy gradient (REINFORCE) over a refined and more compact operation set — comprising depthwise separable convolutions (kernel sizes 3, 5, 7), dilated convolutions (kernel sizes 3, 5), standard convolutions (kernel sizes 3, 5), max pooling, average pooling, and an identity operation — ENAS achieved greater stability under class imbalance. The reward signal explicitly combined AUC and F1 ( $R = 0.6 \times \text{AUC} + 0.4 \times \text{F1}$ ), with a moving-average baseline and

entropy regularisation to encourage architectural diversity. Transitioning from focal loss to binary cross-entropy further stabilised controller learning. ENAS achieved a validation AUC of 0.845 with F1 of 0.62 and recall of 0.964 on the high blood pressure task, demonstrating improved robustness over DARTS under severe class imbalance, but at the cost of longer training and careful reward engineering. Importantly, both DARTS and ENAS operated on seven channels at 128 Hz with a basic temporal representation and twenty disease targets, leaving the multimodal richness of PSG signals substantially underexplored.

The Hierarchical NAS (HNAS) framework was formulated to address the remaining gaps. The search was extended to a nine-channel input with a three-tier feature representation (raw temporal epochs, Welch PSD spectral features, and 96 patient-level clinical statistics), the target set was reduced to six tractable disease conditions to avoid the extreme class imbalance that caused reward oscillation across all twenty conditions, and the search strategy was upgraded from REINFORCE to PPO to provide stable advantage estimation over the noisy zero-cost proxy rewards used for architecture evaluation. The problem is formally stated as: given a single overnight PSG recording comprising nine physiological channels, learn a mapping from multi-stream feature representations to a multi-label binary disease prediction vector for six clinically relevant chronic conditions, under explicit constraints on model parameter count and inference compute, without requiring manual preprocessing or technical expertise from the end user.

## 1.3 Motivation

The primary motivation for this work stems from the convergence of three macro-level trends: the rapid growth of aging populations globally, the increasing clinical evidence linking sleep-disordered physiology to a broad range of chronic conditions, and the recent advances in Neural Architecture Search that make automated lightweight model discovery feasible even under limited compute.

From a clinical standpoint, mental health disorders affect 15–20% of elderly individuals, and cardiovascular conditions represent the leading cause of morbidity and mortality in this demographic. Early and automated detection of conditions such as hypertension, diabetes, and depression through non-invasive sleep recordings could significantly reduce healthcare costs and enable timely interventions before the onset of severe complications.

The three-strategy NAS progression in this work reflects a principled iterative refinement. DARTS established the viability of NAS for PSG health screening, with its reward curve showing steady improvement and convergence by epoch 20 and the discovered architecture outperforming a ResNet-18 baseline at a fraction of its computational cost. ENAS then demonstrated that discrete RL-based search could address the majority-class bias and operational instability of DARTS, achieving a more reliable AUC-F1 trade-off on representative cardiovascular conditions. These two strategies together validated the NAS paradigm for this domain and identified that the key remaining bottleneck was the limitation to a single temporal feature

stream and a flat cell-based search space. HNAS was therefore motivated as the natural extension: a hierarchical search over the full multimodal architecture — CNN feature extractor, sequential encoder, and fusion strategy — evaluated via zero-cost proxy metrics to avoid the noisy reward estimates produced by the three-epoch proxy training used in the initial HNAS variant.

Finally, from a systems perspective, packaging the discovered architecture into a deployable application with plain-language AI-assisted explanation of results directly addresses the accessibility barrier that prevents sleep AI research from reaching elderly patients and their caregivers.

The specific objectives of this project are as follows:

1. To design and investigate multiple Neural Architecture Search frameworks that automatically discover lightweight and efficient architectures for PSG signal analysis on resource-constrained systems — using gradient-based DARTS, RL-based ENAS (REINFORCE), and PPO-based hierarchical HNAS in progressive succession.
2. To develop robust multi-label disease detection models using multimodal PSG signals, progressively improving from seven-channel temporal-only input under DARTS and ENAS to nine-channel three-tier multimodal representation under HNAS, addressing severe class imbalance through asymmetric focal loss and weighted sampling.
3. To design a unified end-to-end application platform (SleepRisk AI) enabling accessible, real-time PSG-based health risk assessment and sleep metric visualisation for non-technical end users.
4. To improve cross-dataset generalisation and reduce demographic bias by integrating large multi-centre PSG cohorts representing heterogeneous elderly populations.

## 1.4 Scope of the Study/Project

and risk prediction

This project focuses on automated multi-disease detection of chronic conditions prevalent in elderly populations using overnight PSG recordings, investigated through three progressive NAS strategies on the CFS dataset.

**DARTS Scope:** Seven physiological channels (EEG C3-M2, EOG, ECG, EMG, thoracic movement, abdominal movement, SpO<sub>2</sub>) standardised to 128 Hz. Twenty disease conditions across cardiovascular, cerebrovascular/mental health, and respiratory categories. A cell-based DAG search space with three stacked cells, two intermediate nodes per cell, and candidate operations including standard convolutions (1×1, 3×1, 5×1), depthwise separable convolutions, dilated convolutions, pooling, two-head attention, and skip connections. Efficiency-aware soft penalties targeting budgets of 50M FLOPs, 25M MACs, and 0.5M parameters. First-order bi-level optimisation with alternating updates. Discovered architecture: 0.17M parameters, 167.8M FLOPs, 82.62M MACs.

**ENAS Scope:** Same seven-channel, 128 Hz, twenty-disease configuration as DARTS. A refined and more compact operation set: depthwise separable convolutions ( $k=3,5,7$ ), dilated convolutions ( $k=3,5$ ), standard convolutions ( $k=3,5$ ), max pooling, average pooling, and identity. An RNN-based RL controller trained via REINFORCE with composite AUC-F1 reward, moving-average baseline, and entropy regularisation. Binary cross-entropy loss replacing focal loss for stable gradient flow. Separate optimizers for shared weights (SGD) and controller (Adam). Achieved validation AUC of 0.845, F1 of 0.62, and recall of 0.964 on high blood pressure.

**HNAS Scope:** Nine channels (adding airflow and nasal pressure) at 100 Hz. Six disease conditions: depression, high blood pressure, diabetes, asthma, hay fever, and sinusitis. Three complementary feature streams: raw temporal 30-second epochs, 45 spectral features per window via Welch’s PSD, and 96 patient-level clinical features. A hierarchical search space of approximately 30 sequential decision tokens covering CNN depth, width, per-layer operation, batch normalisation, pooling, encoder type, encoder hyperparameters, cross-modal attention, clinical injection stage, and fusion strategy. PPO-trained two-layer LSTM controller (hidden size 256) with zero-cost proxy evaluation combining NASWOT, FreqCV, SeqFlow, and Class-Gap scores. Discovered architecture: 3-layer CNN ( $\text{dil}3 \rightarrow \text{dil}7 \rightarrow \text{res}5$ ), TCN temporal encoder, GRU spectral encoder, sum fusion, late clinical injection; 0.386M parameters, 347.31 MFLOPs; macro AUC-ROC 0.6676. Desktop deployment application (SleepRisk AI) with full inference pipeline, AI chatbot, and automated PDF medical reports.

**Exclusions:** This project does not include prospective clinical validation on live patient data, real-time hardware integration with PSG acquisition devices, or regression-style continuous risk score prediction. Integration of additional multi-centre cohorts (SHHS, MESA, WSC, MrOS) is identified as future work.

**Report Organisation:** The remainder of this report is organised as follows. Chapter 2 reviews existing literature and tools relevant to PSG-based automated disease detection, identifies research gaps, and formally states the problem and contributions across all three NAS strategies. Chapter 3 describes the dataset, preprocessing pipeline, and three-tier feature representation strategy. Chapter 4 presents the DARTS and ENAS frameworks, including their respective search spaces, optimisation strategies, and results. Chapter 5 presents the HNAS framework, the PPO controller, and zero-cost proxy evaluation. Chapter 6 reports experimental results including classification performance, ablation studies, and comparison with state-of-the-art models. Chapter 7 describes the SleepRisk AI application platform. Chapter 8 concludes with a summary of findings and directions for future work.

# Chapter 2

## Literature Review/ Existing System

### 2.1 Existing System Study

#### 2.1.1 Review of Existing Research Work

The automated analysis of polysomnography data for clinical prediction has evolved from classical machine learning approaches towards large-scale deep learning and foundation model paradigms. This section reviews representative prior work organised by methodological lineage.

**Classical Feature-Based Approaches.** Early work on automated PSG analysis relied on hand-crafted feature extraction followed by conventional classifiers. Fallmann and Chen (2018) proposed a system for detecting chronic diseases including hypertension, diabetes, and sleep apnea using sleep-wake behaviour and clinical features extracted from PSG signals. While the approach was non-invasive and suitable for long-term monitoring outside clinical settings, it was limited to simple single-modality features and achieved substantially lower accuracy than modern deep learning methods, with reported accuracy in the range of 0.62–0.81 depending on the condition. Blanchard et al. (2021) developed an AI-based system for six-year cardiovascular morbidity and all-cause mortality prediction in patients suspected of sleep apnea using physiological measurements from the ERMES study group. Their approach was lightweight and interpretable but relied on shallow feature engineering and may miss subtle neurological or fine-grained physiological patterns captured by deep learning on full waveforms.

**Convolutional and Recurrent Deep Learning Models.** Zhang et al. (2024) proposed a multi-modal sleep stage classification architecture using a two-stream encoder-decoder design with separate CNN branches for EEG and EOG signals combined with a BiGRU decoder, published in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. The model achieved competitive staging accuracy on the SHHS dataset but was designed exclusively for sleep staging rather than disease-level prediction. Liu et al. (2025) introduced MHFNet, a multimodal hybrid-embedding fusion network for automatic sleep staging combining convolutional feature extraction with transformer-based temporal modelling, published in the *IEEE Journal of Biomedical and Health Informatics*. While achieving strong staging performance, the architecture was not evaluated for multi-disease chronic condition detection and remains



computationally demanding for edge deployment. Lee et al. (2025) presented an explainable vision transformer for automatic sleep staging on multimodal PSG signals in *npj Digital Medicine*, using attention maps for visual interpretability. The transformer backbone incurred high computational cost relative to lightweight CNN-based alternatives.

**Self-Supervised and Contrastive Pretraining.** Jiang et al. (2021) applied self-supervised contrastive learning to EEG-based sleep staging at IJCNN, demonstrating that label-efficient pretraining can significantly improve staging accuracy under low-annotation regimes. Weng et al. (2025) provided a comprehensive survey of self-supervised learning methods for EEG in *ACM Computing Surveys*, highlighting the growing utility of representation learning for biosignal analysis. These approaches are primarily validated on EEG staging tasks and have not been extended to multimodal PSG-based chronic disease detection.

**Cardiovascular Risk Profiling from PSG.** He et al. (2025) proposed a multimodal cardiovascular risk profiling system using PSG signals with self-supervised learning, published in *Sleep Journal*, targeting outcomes such as CVD mortality, incident hypertension, and atrial fibrillation. Their hybrid ResNet-Transformer architecture generated interpretable risk scores from multimodal PSG data and eliminated the need for manual sleep staging, but was not optimised for lightweight or edge deployment. Kim et al. (2025) presented a multimodal AI system for automated screening of cardiovascular diseases using nocturnal physiological signals in *npj Cardiovascular Health*, achieving strong sensitivity for cardiac conditions but again relying on computationally heavy architectures. Manimaran et al. (2025) proposed a cardiovascular risk assessment system combining sleep patterns with ECG-based biological age estimation in the *Journal of Clinical Medicine*, providing interpretable risk profiles using ECG-derived features alongside sleep metrics.

**Sleep Foundation Models.** Two large-scale foundation model efforts represent the current state of the art. Thapa et al. (2025) proposed SleepFM, a multimodal sleep foundation model trained on over 500,000 hours of PSG recordings from approximately 65,000 participants. SleepFM is channel-agnostic and demonstrated prediction capability for more than 130 conditions including heart failure, dementia, stroke, and chronic kidney disease. However, SleepFM requires massive pretraining compute and large proprietary datasets, making it infeasible for deployment in resource-constrained settings. At 4.44 million parameters, SleepFM is approximately 26 times larger than the DARTS/ENAS-discovered architecture and approximately 11.6 times larger than the HNAS-discovered architecture. Nie et al. (2025) introduced a zero-burden sleep foundation model built on cardiorespiratory signals from over 800,000 hours of multi-ethnic sleep recordings targeting conditions including Parkinson’s disease, GERD, heart failure, CVD death, and sleep apnea. Despite its accessibility potential through home-based BCG sensors, the model relies on a complex hierarchical ResNet-Transformer architecture that is computationally heavy for low-end or mobile deployment.

Bilal et al. (2025) proposed a foundation model for sleep-based risk stratification and clinical outcomes evaluated on cardiological endpoints using the MESA dataset. Kjaer et al. (2025) presented the Stanford Sleep Bench for evaluating polysomnography pre-training methods for

sleep foundation models, providing a systematic benchmark for comparing representation learning strategies across PSG tasks.

### 2.1.2 Review of Existing Tools/Products

Several commercial and open-source tools exist in the PSG analysis and sleep medicine software landscape, but most fall short of providing automated multi-disease risk profiling from raw PSG signals.

**Natus SleepWorks and Compumedics Profusion:** Industry-standard clinical PSG analysis platforms used in hospital sleep labs. They provide automated sleep staging assistance and respiratory event detection using proprietary rule-based algorithms but require expensive hardware infrastructure, do not provide chronic disease risk prediction, and are not intended for resource-limited deployment.

**YASA (Yet Another Spindle Algorithm):** An open-source Python library providing automated sleep staging from EEG signals using LightGBM classifiers with hand-crafted features. YASA operates on single EEG channels and does not support multimodal PSG input, chronic disease classification, or patient-level risk aggregation.

**SleepEDF Toolbox and MNE-Python:** Open-source research toolkits for PSG signal loading, preprocessing, and visualisation. They provide no built-in disease prediction capability and require substantial custom development for clinical deployment.

**SleepFM (Thapa et al., 2025):** As reviewed above, SleepFM represents the closest prior work to a unified PSG-based multi-disease prediction system. With 4.44 million parameters and requiring large-scale pretraining across 65,000+ subjects, it is not practically deployable in resource-constrained settings.

**TinySleepNet:** A compact CNN-LSTM architecture proposed for EEG-based sleep staging operating on a single EEG channel with 0.74 million parameters. TinySleepNet demonstrates the feasibility of lightweight architecture design for sleep biosignals but does not extend to multimodal PSG input or chronic disease classification.

In summary, no existing commercial tool or published open-source framework simultaneously satisfies the requirements of multimodal PSG processing, multi-disease chronic condition detection, lightweight deployability, and an accessible end-user interface for non-technical users. The proposed SleepRisk AI system, developed through the three-strategy NAS progression, addresses all four requirements within a unified framework.

## 2.2 Research Gap / Scope for Improvement and Innovation

The preceding review reveals several persistent limitations in existing systems and research:

1. **Computational Weight vs. Deployment Constraint Mismatch:** Existing high-performing PSG-based disease prediction models (SleepFM, ResNet-Transformer hybrids,

SleepFounder) are computationally heavy, requiring substantial GPU resources for both training and inference. This limits their deployment on low-resource or edge healthcare devices commonly available in rural clinical settings.

2. **Single-Disease Focus:** The majority of published PSG AI systems target a single clinical outcome — most commonly sleep staging or obstructive sleep apnea severity — leaving the rich diagnostic content of PSG signals untapped for simultaneous multi-disease chronic condition prediction in elderly populations.
3. **Incomplete Multimodal Utilisation:** Current approaches typically process only a subset of available PSG channels. Cross-modal diagnostic information spanning respiratory, cardiac, cortical, and movement channels remains underexplored in unified prediction frameworks.
4. **NAS Unexplored for PSG Health Screening:** Prior to this work, Neural Architecture Search and model compression techniques had not been applied to PSG-based chronic disease detection. The accuracy-efficiency trade-off for this domain was therefore entirely unoptimised.
5. **Class Imbalance Under Multi-label Disease Settings:** Gradient-based NAS objectives that maximise overall accuracy produce architectures biased towards majority disease classes, resulting in near-zero recall for rare conditions. This was directly observed with DARTS, and partially addressed by ENAS through explicit AUC-F1 reward formulation before being more fully resolved in HNAS through asymmetric focal loss and the ClassGap proxy penalty.
6. **Operational Bias in Gradient-Based Search:** DARTS exhibited a well-known tendency to favour skip connections and pooling operations over more expressive convolutional operations, reducing the effective representational depth of discovered architectures. This motivated the shift to discrete RL-based search in ENAS, which samples concrete operations directly.
7. **Absence of a Unified Multi-Disease Detection Framework:** No single end-to-end framework exists that simultaneously detects cardiovascular, respiratory, and neuropsychiatric conditions from a single overnight PSG recording, which is essential for cost-efficient screening in elderly cohorts with high comorbidity rates.
8. **Lack of Accessible Deployment Systems:** No existing sleep AI solution provides a complete, accessible end-user application capable of accepting raw PSG files, running inference without manual preprocessing, and presenting results in clinician-readable format.

**Gaps addressed by DARTS:** DARTS addressed gaps (1), (2), and (4) by demonstrating, for the first time, that a gradient-efficient NAS framework could discover a lightweight cell-based

architecture (0.17M parameters, 167.8M FLOPs, 82.62M MACs) for PSG multi-label disease classification. The discovered model achieved mean AUC-ROC of 0.845, 0.769, and 0.699 across cardiovascular, mental health, and respiratory disease cohorts respectively. However, DARTS exposed gaps (5) and (6): majority-class optimisation bias and operational collapse toward skip connections.

**Gaps addressed by ENAS:** ENAS addressed gaps (5) and (6) by replacing gradient-based search with a discrete RL controller (REINFORCE) trained on a composite AUC-F1 reward with moving-average baseline and entropy regularisation. The refined operation set — narrower and better-suited for discrete sampling than the DARTS space — eliminated the skip-connection collapse problem. Transitioning from focal loss to binary cross-entropy provided smoother gradient flow for the shared supernet. ENAS achieved a validation AUC of 0.845, F1 of 0.62, and recall of 0.964 on high blood pressure classification, demonstrating improved robustness over DARTS under severe class imbalance. However, both DARTS and ENAS operated on a single temporal feature stream with seven channels, leaving gaps (3) and (7) unresolved.

**Gaps addressed by HNAS:** HNAS addressed gaps (3), (7), and (8). The extension to nine channels with a three-tier multimodal feature representation (temporal, spectral, and clinical) addressed incomplete modality utilisation. The hierarchical search over macro-level structural decisions — CNN depth and width, encoder type, cross-modal attention, clinical injection stage, and fusion strategy — alongside micro-level per-layer operation choices constituted the first unified multi-disease prediction framework from PSG under a lightweight compute budget. The SleepRisk AI deployment application addressed gap (8) through an accessible end-to-end platform. The reduction to six tractable disease conditions addressed the extreme class imbalance that caused persistent reward oscillation across all twenty conditions in the earlier strategies.

**Remaining gaps for future work:** Full edge hardware deployment on embedded systems, multi-centre cross-cohort generalisation at scale (SHHS, MESA, WSC, MrOS), and regression-style disease risk score prediction using literature-validated hazard ratios remain as planned future work.

## 2.3 Problem Statement and Contributions

1. **Problem Statement:** Manual PSG analysis is slow, expensive, and inaccessible in resource-limited healthcare settings. Existing deep learning solutions are computationally intensive, limited to single-disease prediction, and fail to generalise across diverse elderly populations. Neural Architecture Search strategies for this domain are unexplored and the accuracy-efficiency trade-off is unoptimised. There is therefore a need to develop a *lightweight, generalizable, and multimodal deep learning framework for automated PSG analysis* that can simultaneously detect multiple chronic conditions prevalent in elderly populations without requiring high-end hardware or technical expertise from the end

user. Formally, given a single overnight PSG recording, the system must learn a mapping from multi-stream feature representations to a multi-label binary disease prediction vector covering clinically relevant chronic conditions, while satisfying explicit constraints on model parameter count and inference compute budget, under severe class imbalance conditions.

## 2. Research Contributions:

- (a) **First Application of Gradient-Based NAS to PSG Multi-Disease Health Screening (DARTS):** A DARTS-based gradient-efficient NAS framework was applied to discover lightweight cell-based architectures for PSG multi-label disease classification, using a cell structure with three stacked cells, two intermediate nodes, and a rich candidate operation set subject to efficiency-aware soft constraints targeting 50M FLOPs, 25M MACs, and 0.5M parameters. The discovered architecture (0.17M parameters, 167.8M FLOPs, 82.62M MACs) achieved AUC-ROC of 0.845, 0.769, and 0.699 across cardiovascular, mental health, and respiratory disease cohorts respectively. This constitutes the first use of gradient-based NAS for PSG-based chronic disease detection.
- (b) **Reinforcement Learning-Based Discrete Architecture Search for Robust Imbalanced PSG Classification (ENAS):** ENAS was investigated as a discrete RL-based NAS strategy to address the majority-class bias and skip-connection collapse observed with DARTS. A recurrent controller trained via REINFORCE with a composite AUC-F1 reward ( $R = 0.6 \times \text{AUC} + 0.4 \times \text{F1}$ ), moving-average baseline, and entropy regularisation explored a refined operation set better suited to discrete sampling. Transitioning from focal loss to BCE for shared supernet training stabilised controller learning dynamics. The ENAS-derived architecture achieved a validation AUC of 0.845 with F1 of 0.62 and recall of 0.964 on high blood pressure classification, demonstrating improved robustness over DARTS under severe class imbalance. This represents the first application of ENAS-style discrete RL architecture search to multivariate PSG health risk classification.
- (c) **Hierarchical RL-Based NAS with Zero-Cost Proxy Evaluation (HNAS):** A novel PPO-driven hierarchical NAS framework is proposed in which a two-layer LSTM controller (hidden size 256) discovers architectures over approximately 30 sequential decision tokens spanning macro-level structural choices (CNN depth, width, skip connectivity, encoder type, cross-modal attention, clinical injection stage, fusion strategy) and micro-level per-layer operation choices. Architecture evaluation uses a composite zero-cost proxy combining NASWOT (activation expressivity, weight 0.5), FreqCV (frequency selectivity, weight 0.3), SeqFlow (temporal gradient propagation, weight 0.2), and ClassGap (majority class penalty,  $-0.1$ ), with a receptive field validity gate that rejects architectures before any model construction. Each candidate is evaluated in 3–8 seconds without any gradient descent. This eliminates

the noisy shared-weight proxy training used in the initial HNAS variant and, to the best of our knowledge, represents the first application of PPO-based hierarchical NAS to multimodal PSG disease prediction.

- (d) **Lightweight HNAS-Discovered Multimodal Architecture:** The PPO search discovered a 3-layer dilated CNN (dil3→dil7→res5) coupled with a TCN temporal encoder, a GRU spectral encoder, and sum fusion with late clinical feature injection. With 385,734 parameters (0.386M) and 347.31 MFLOPs, the architecture achieves a macro AUC-ROC of 0.6676 across six disease conditions, matching SleepFM’s performance on depression (0.71 vs. 0.74) and equalling it on diabetes (0.72 vs. 0.72), while training on 619 subjects versus SleepFM’s 65,000+ at approximately 11.6-fold lower parameter count.
- (e) **Three-Tier Multimodal Feature Representation:** A principled three-stream feature representation is introduced combining raw temporal signals (9 channels, 30-second epochs at 100 Hz), Welch PSD-derived spectral features (45 features per overlapping window), and 96 patient-level aggregate clinical features (sleep architecture, respiratory events, HRV, EEG spectral indices). Ablation studies confirm that the multimodal combination (macro AUC-ROC 0.6676) outperforms temporal-only (0.4198), spectral-only (0.6500), and clinical-only (0.6408) baselines.
- (f) **Accessible End-to-End Deployment Platform — SleepRisk AI:** A complete desktop application is developed that accepts raw EDF and XML PSG files and runs the full inference pipeline end-to-end without manual preprocessing. The platform outputs risk percentages for six disease conditions alongside interpretable sleep architecture metrics, provides AI-assisted result explanation via an integrated LLM chatbot (Sleep Copilot), and generates structured clinician-readable PDF medical reports, directly addressing the clinical accessibility gap identified in the literature.

# Chapter 3

## Proposed Work

### 3.1 Proposed Work

The preceding literature review establishes that while deep learning has demonstrated significant promise for automated PSG analysis, existing solutions suffer from a fundamental tension between predictive performance and practical deployability. State-of-the-art systems such as SleepFM and the SleepFounder model rely on architectures with millions of parameters trained on hundreds of thousands of hours of data, placing them far beyond the computational reach of resource-limited healthcare settings that serve elderly populations most in need of continuous health monitoring. Simultaneously, prior approaches are predominantly single-disease in scope, process only a subset of available PSG modalities, and have never been subjected to Neural Architecture Search techniques that could automatically discover efficient architectures for this domain.

This chapter presents the proposed *SleepRisk AI* system — a lightweight, multimodal, and accessible framework for automated chronic disease risk assessment from overnight PSG recordings. The proposed work directly addresses the eight research gaps identified in Chapter 2 through a principled three-stage NAS progression: Differentiable Architecture Search (DARTS), Efficient Neural Architecture Search (ENAS), and a novel Hierarchical Neural Architecture Search (HNAS) framework driven by Proximal Policy Optimisation (PPO). Each strategy builds on the limitations of its predecessor, culminating in a discovered architecture that simultaneously detects six chronic conditions prevalent in elderly populations using nine-channel multimodal PSG input while remaining deployable on standard consumer hardware.

#### 3.1.1 Objectives of the Proposed Work

The following objectives were defined to guide the investigation:

1. **Primary Objective:** To design and investigate multiple Neural Architecture Search frameworks that automatically discover lightweight and efficient deep learning architectures for PSG-based multi-disease health risk classification on resource-constrained systems, progressing from gradient-based DARTS through RL-based ENAS to PPO-driven

hierarchical HNAS in successive stages of refinement.

2. **Secondary Objective 1:** To develop robust multi-label disease detection models using multimodal PSG signals, progressively improving from a seven-channel temporal-only representation under DARTS and ENAS to a nine-channel three-tier multimodal representation under HNAS, addressing severe class imbalance through asymmetric focal loss, weighted sampling, and explicit minority-class-aware proxy metrics.
3. **Secondary Objective 2:** To design and implement a unified end-to-end application platform (SleepRisk AI) enabling accessible, real-time PSG-based health risk assessment and sleep metric visualisation for non-technical end users, without requiring any manual preprocessing or domain expertise.
4. **Secondary Objective 3:** To lay the groundwork for improved cross-dataset generalisation and reduced demographic bias through a preprocessing and feature representation pipeline designed to be compatible with large multi-centre PSG cohorts, with integration of SHHS, MESA, WSC, and MrOS identified as immediate future work.

Collectively, these objectives define a measurable research programme with clear deliverables: a lightweight NAS-discovered architecture benchmarked against state-of-the-art models on AUC-ROC, F1-score, and average precision metrics; an ablation study validating the contribution of each feature stream; and a functional deployment application validated on real CFS recordings.

## 3.2 Methodology

### 3.2.1 Overview of the Approach

The proposed methodology is structured as a four-module end-to-end pipeline. The first module covers dataset curation and preprocessing, transforming raw EDF-format PSG recordings into a three-tier tensor representation comprising temporal, spectral, and clinical feature streams. The second module defines the hierarchical search space over which NAS operates, spanning both micro-level operation choices within convolutional layers and macro-level structural decisions governing encoder type, cross-modal interaction, and fusion strategy. The third module encompasses the three NAS search strategies — DARTS, ENAS, and HNAS — that progressively refine the architecture discovery process. The fourth module covers full model training, evaluation, and deployment via the SleepRisk AI application.

Figure 3.1 illustrates the complete three-phase methodology pipeline. Phase 1 performs architecture search using the NAS framework. Phase 2 trains the discovered architecture end-to-end on the full dataset using the three-tier feature representation. Phase 3 deploys the trained model in a desktop application that accepts raw PSG files and produces clinician-readable risk reports.



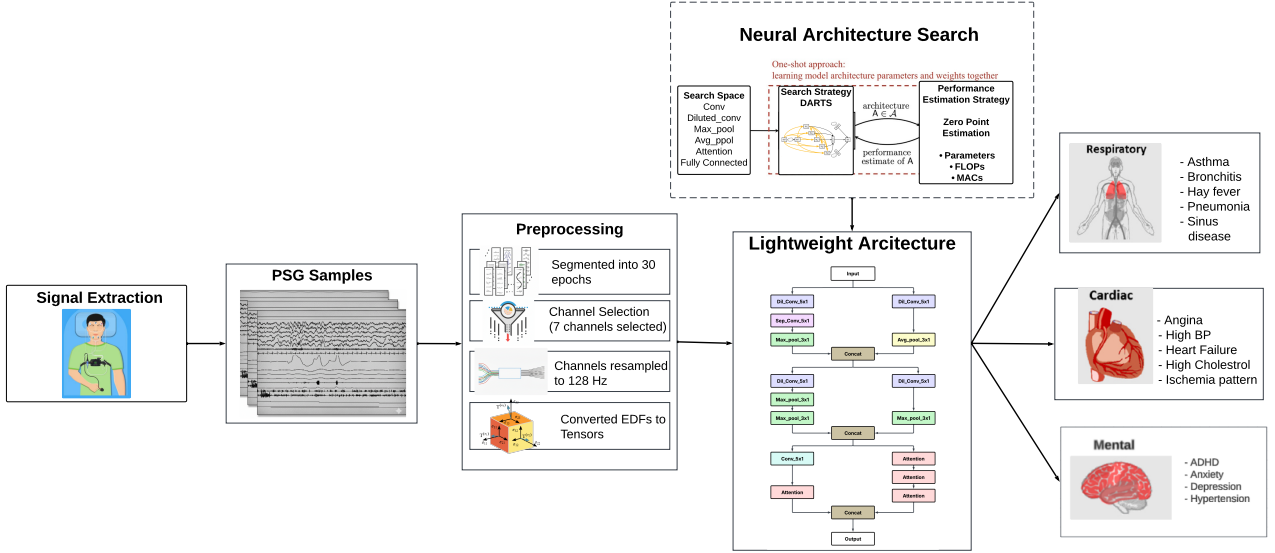


Figure 3.1: Three-phase methodology pipeline: (1) Preprocessing of the PSG signals, (2) Architecture Search via DARTS/ENAS/HNAS, (3) Full model training and classification of diseases

### 3.2.2 Dataset Selection

#### Dataset Selection and Justification

The Cleveland Family Study (CFS) Sleep dataset was selected as the primary dataset for this investigation. The CFS was accessed through the National Sleep Research Resource (NSRR) and represents a large family-based cohort originally designed to study the epidemiology and risk factors of sleep-disordered breathing, with a particular focus on obstructive sleep apnea. Initiated in 1990, the study enrolled 2,284 individuals drawn from 361 families, with approximately 46% of participants identifying as African American — making the CFS one of the more ethnically diverse PSG cohorts available for research. Subjects were assessed on up to four occasions over a longitudinal period spanning nearly 16 years, with the final examination conducted in February 2006.

The dataset includes overnight PSG recordings from 730 individuals which were used in this study. Recordings were performed using a standardized 14-channel acquisition protocol capturing multichannel EEG, EOG, EMG, oronasal airflow, thoracic and abdominal respiratory effort, oxygen saturation, and ECG signals. Signals were acquired at modality-specific sampling rates ranging from 1 to 512 Hz. Trained technicians performed sensor calibration and impedance checks prior to each recording, and all nocturnal recordings were transmitted to a centralized reading center at Brigham and Women’s Hospital where sleep stages and respiratory events were manually scored according to current clinical guidelines.

Among the PSG datasets surveyed — including CFS, SHHS, MESA, WSC, and MrOS — CFS was selected for initial experimentation for two reasons. First, its size (730 recordings after filtering) provides a tractable training set for NAS-based architecture discovery without requiring distributed training infrastructure. Second, CFS contains all nine physiological channels required by the full HNAS input representation, as well as rich disease annotation from

clinical evaluations and medical records, making it suitable for the multi-label chronic disease classification task.

## Disease Annotation and Target Selection

Health risk annotations were derived from clinical evaluations and medical records in the CFS dataset. All disease records were first reviewed to obtain a comprehensive understanding of available labels. Diseases were grouped into three cohorts closely associated with elderly populations: cardiovascular, respiratory, and cerebrovascular/mental health conditions.

A sample count filter was applied to retain only conditions with sufficient prevalence for reliable training. This initial filter yielded twenty conditions spanning the three cohorts, categorized as follows.

- **Cardiovascular (9 conditions):** Myocardial infarction, angina, high blood pressure, cardiovascular disease, high cholesterol, congestive heart failure, irregular heartbeat, stroke, ischemia pattern.
- **Cerebrovascular and Mental Health (6 conditions):** Cerebrovascular disease, ADHD, anxiety, behavior disorder, depression, hypertension.
- **Respiratory (5 conditions):** Asthma, bronchitis, hay fever, pneumonia, sinus disease.

For the DARTS and ENAS search experiments, all twenty conditions were retained as classification targets. For the HNAS experiments, it was observed that several of the twenty conditions exhibited extreme class imbalance with positive sample rates as low as 2–5% of the total population. This level of imbalance introduced persistent reward oscillation and unreliable gradient dynamics during both architecture search and full model training. After systematic analysis of label prevalence and training stability across all conditions, the target set for HNAS was reduced to six conditions that offered a more tractable distribution: depression, high blood pressure, diabetes, asthma, hay fever, and sinusitis. This reduction was not motivated by clinical irrelevance but by the practical necessity of achieving reliable convergence within the data and computational constraints of the study.

The problem is formulated as a multi-label classification task where each subject may present multiple concurrent comorbidities, reflecting real-world clinical complexity.

## Selected PSG Channels

Channel selection was guided by the objective of developing a lightweight yet physiologically comprehensive pipeline for multivariate time-series health risk classification. For the DARTS and ENAS configurations, seven channels were selected: ECG, EMG, thoracic movement, abdominal movement, SpO<sub>2</sub>, EOG, and a bipolar EEG derivation (C3-M2). For the HNAS configuration, two additional channels — nasal pressure and airflow — were incorporated to give a nine-channel input providing richer multimodal coverage of cardiovascular, respiratory, and neurological processes.

Table 3.1 summarises the selected channels and their clinical relevance.

Table 3.1: Selected PSG channels and their clinical significance.

| Channel          | Physiological Domain            | NAS Config      |
|------------------|---------------------------------|-----------------|
| EEG (C3-M2)      | Brain activity, sleep stages    | DARTS/ENAS/HNAS |
| EOG              | Eye movements, REM detection    | DARTS/ENAS/HNAS |
| ECG              | Cardiac rhythm, HRV             | DARTS/ENAS/HNAS |
| EMG              | Muscle tone, respiratory effort | DARTS/ENAS/HNAS |
| Thoracic Effort  | Breathing mechanics             | DARTS/ENAS/HNAS |
| Abdominal Effort | Diaphragm activity              | DARTS/ENAS/HNAS |
| SpO <sub>2</sub> | Blood oxygen saturation         | DARTS/ENAS/HNAS |
| Airflow          | Upper-airway flow               | HNAS only       |
| Nasal Pressure   | Upper-airway dynamics           | HNAS only       |

### 3.2.3 Algorithm/Model Design

#### Data Preprocessing Pipeline

The preprocessing pipeline transforms raw EDF-format PSG recordings into structured tensor representations. The pipeline consists of the following sequential steps.

**Removal of Pre-Sleep Wake Segments.** Each recording’s XML hypnogram was parsed to identify scored sleep stages. The initial wake segment preceding the first annotated sleep-stage transition was detected and removed, so that model inputs reflect only physiologically meaningful sleep rather than pre-sleep wakefulness. This step increases consistency between subjects and reduces noise from non-informative segments.

**Bipolar EEG Construction (DARTS and ENAS).** For the DARTS and ENAS configurations, a single bipolar EEG derivation (C3-M2) was formed by computing the differential between the C3 and M2 electrode signals. This minimises reference-related artifacts. Missing channels were handled through zero-padding to maintain a fixed channel count across all recordings.

**Signal Smoothing and Artefact Reduction (HNAS).** For the HNAS configuration, a Savitzky-Golay filter was applied to each of the nine channels after wake removal. The filter smooths the signals while preserving physiologically relevant transients and the underlying morphological structure, which is particularly important for the EEG and respiratory channels.

**Resampling.** All channels were resampled to a uniform target sampling frequency. DARTS and ENAS configurations use 128 Hz to preserve the full spectral content of EEG and EMG signals. The HNAS configuration uses 100 Hz, which provides sufficient resolution for all clinically relevant sleep oscillations while reducing the per-epoch sample count from 3840 to 3000 samples, easing the computational load during CNN-based epoch feature extraction.

**Channel-Wise Normalisation.** Z-score normalisation was applied independently to each channel:

$$x' = \frac{x - \mu}{\sigma} \quad (3.1)$$

where  $\mu$  and  $\sigma$  denote the channel-specific mean and standard deviation computed over the training set. This ensures numerical stability and prevents high-amplitude modalities from dominating the learning process.

**Fixed-Length Alignment and Tensor Conversion.** For DARTS and ENAS, each multichannel signal was aligned to a fixed temporal length via linear interpolation, producing a uniform tensor of shape  $C \times T$ . For HNAS, each recording was segmented into consecutive non-overlapping 30-second epochs, each stored as a slice of shape  $C \times T_{\text{epoch}}$  where  $T_{\text{epoch}} = 3000$  samples at 100 Hz. EDF files were serialized to PyTorch tensor format, reducing I/O overhead during training by 50–100× compared to repeated EDF parsing.

Figure 3.2 illustrates the HNAS preprocessing pipeline from raw EDF input to tensor-formatted epochs.

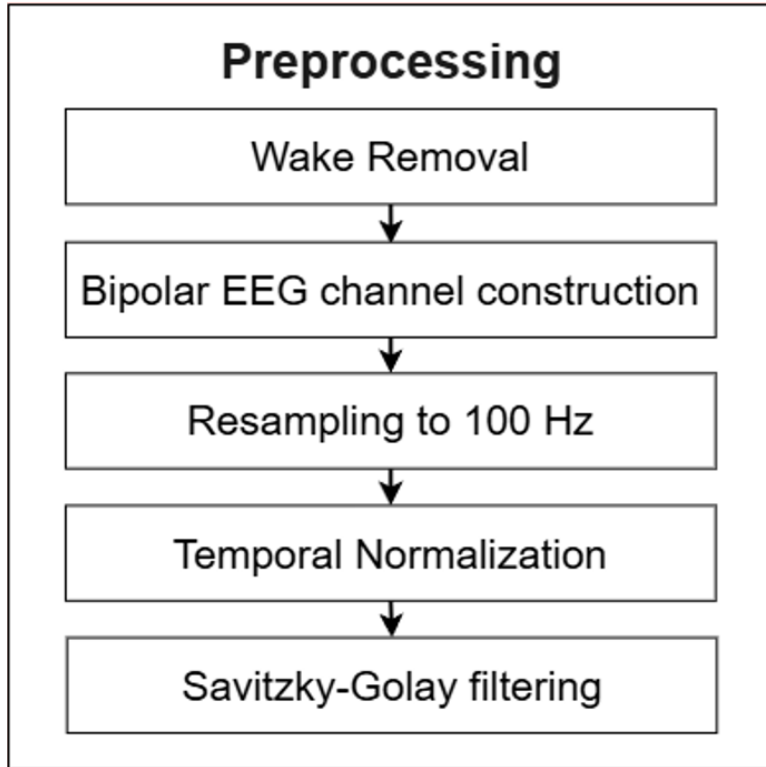


Figure 3.2: HNAS preprocessing pipeline. Steps proceed from top to bottom: wake removal using hypnogram annotations, bipolar EEG construction, resampling to 100 Hz, temporal normalisation, Savitzky-Golay smoothing, and tensor serialisation. Each patient’s recording is stored as a tensor of shape  $[X, 9, 3000]$  where  $X$  is the number of sleep epochs.

### Three-Tier Feature Representation

A central contribution of this work is a principled three-stream feature representation designed to capture complementary dimensions of the PSG signal. The motivation is that PSG signals

carry information along two orthogonal dimensions — temporal dynamics and spectral content — which a single representation cannot capture simultaneously. A third stream provides patient-level aggregate context that no epoch-level stream can encode. The three streams are described below.

**Stream 1: Temporal Features.** The raw nine-channel PSG signal is sampled at 100 Hz, producing 3000 data points per 30-second epoch. Recordings are segmented into non-overlapping 30-second windows and reshaped to  $[X, 9, 3000]$  per patient, where  $X$  is the number of sleep epochs in the night. Each patient constitutes one sample. A batch of  $B$  patients is padded to  $[B, X_{\max}, 9, 3000]$  with a validity mask per patient to handle variable recording lengths. The epoch-sequential structure preserves night-long temporal order, enabling the sequence encoder to learn sleep-stage transitions, respiratory event patterns, and other temporal dynamics that unfold across the night.

**Stream 2: Spectral Features.** Thirty-second windows with 50% overlap are extracted from the nine-channel PSG signal. For each window, Welch’s power spectral density (PSD) is computed independently per channel, and a set of frequency-domain features are extracted as summarised in Table 3.2. A total of 45 features are extracted per window per patient, producing a feature matrix of shape  $[B, W, 45]$  where  $W$  is the number of overlapping windows across the night. These features are stored as PyTorch tensors aligned with the temporal stream for joint multimodal processing.

Table 3.2: Spectral features extracted per window per channel.

| Channel                     | Features Extracted                                                                                              |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| EEG                         | Absolute & relative band power ( $\delta/\theta/\alpha/\sigma/\beta/\gamma$ ), spectral entropy, peak frequency |
| EOG                         | Relative band power ( $\delta/\theta/\alpha/\sigma/\beta/\gamma$ ), spectral entropy                            |
| ECG                         | LF power, HF power, LF/HF ratio, spectral entropy                                                               |
| EMG                         | Band power (10–30 Hz, 30–60 Hz), peak frequency, spectral entropy                                               |
| Respiratory<br>(4 channels) | Peak frequency, respiratory band power (0.1–0.5 Hz), spectral entropy                                           |
| SpO <sub>2</sub>            | VLF & LF band power, spectral entropy, signal variance                                                          |

**Stream 3: Clinical Features.** Patient-level aggregate features are extracted from the full overnight PSG recording using sleep stage annotations. A total of 96 features are computed, spanning sleep architecture statistics (total sleep time, wake after sleep onset, sleep stage percentages), respiratory event indices (apnea-hypopnea index, respiratory disturbance index, oxygen desaturation index), heart rate variability metrics, EEG spectral band indices, and cross-signal coupling measures. These clinical features provide patient-level physiological context that neither the temporal nor the spectral stream can encode at the epoch level. Z-score normalisation is applied using training-set statistics, and feature values are clipped to  $\pm 5$  to suppress outlier effects. Clinical features are injected into the model at a stage determined by

the PPO architecture search (early injection before the sequence encoder or late injection at the fusion stage).

Figure 3.3 illustrates the three-stream feature representation.

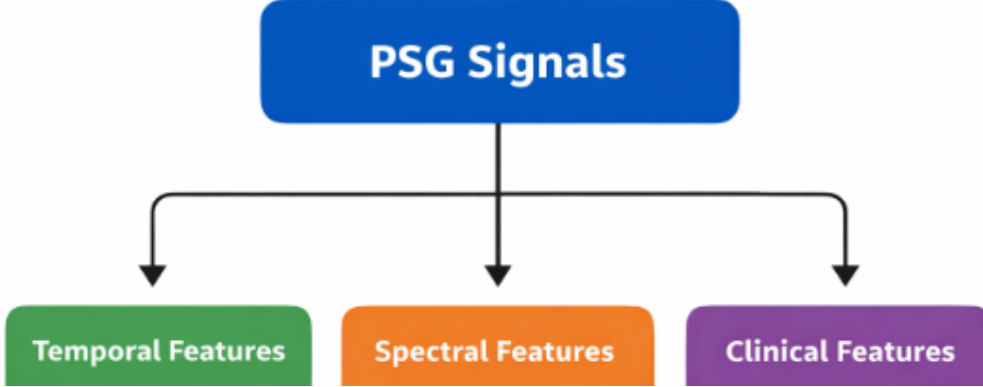


Figure 3.3: Three-tier feature representation extracted from PSG signals. The temporal stream provides raw epoch-level dynamics, the spectral stream provides frequency-domain characteristics, and the clinical stream provides patient-level aggregate physiological context. All three streams are jointly processed by the HNAS-discovered model.

### Neural Architecture Search

Neural Architecture Search (NAS) is an automated method for discovering optimal neural network architectures by navigating a defined search space using a search strategy, evaluated through a performance estimation strategy. Rather than relying on manually designed architectures and post-hoc compression, NAS formulates architecture design as an optimization problem, enabling joint exploration of model structure and performance under explicit computational constraints.

Three NAS strategies are investigated in this work, each characterized by three fundamental components: the *search space* (the set of candidate operations and structural building blocks), the *search strategy* (the method for exploring candidate architectures), and the *evaluation measure* (the technique for assessing candidate quality without full training).

**DARTS: Differentiable Architecture Search.** DARTS introduces a continuous relaxation over discrete architectural choices by representing each edge in a directed acyclic graph (DAG) as a softmax-weighted mixture of candidate operations, enabling gradient-based joint optimisation of network weights and architecture parameters. For an edge  $(i, j)$  in the DAG, the mixed operation is:

$$\tilde{o}^{(i,j)}(x) = \sum_{o \in \mathcal{O}} \frac{\exp(\alpha_o^{(i,j)})}{\sum_{o' \in \mathcal{O}} \exp(\alpha_{o'}^{(i,j)})} \cdot o(x) \quad (3.2)$$

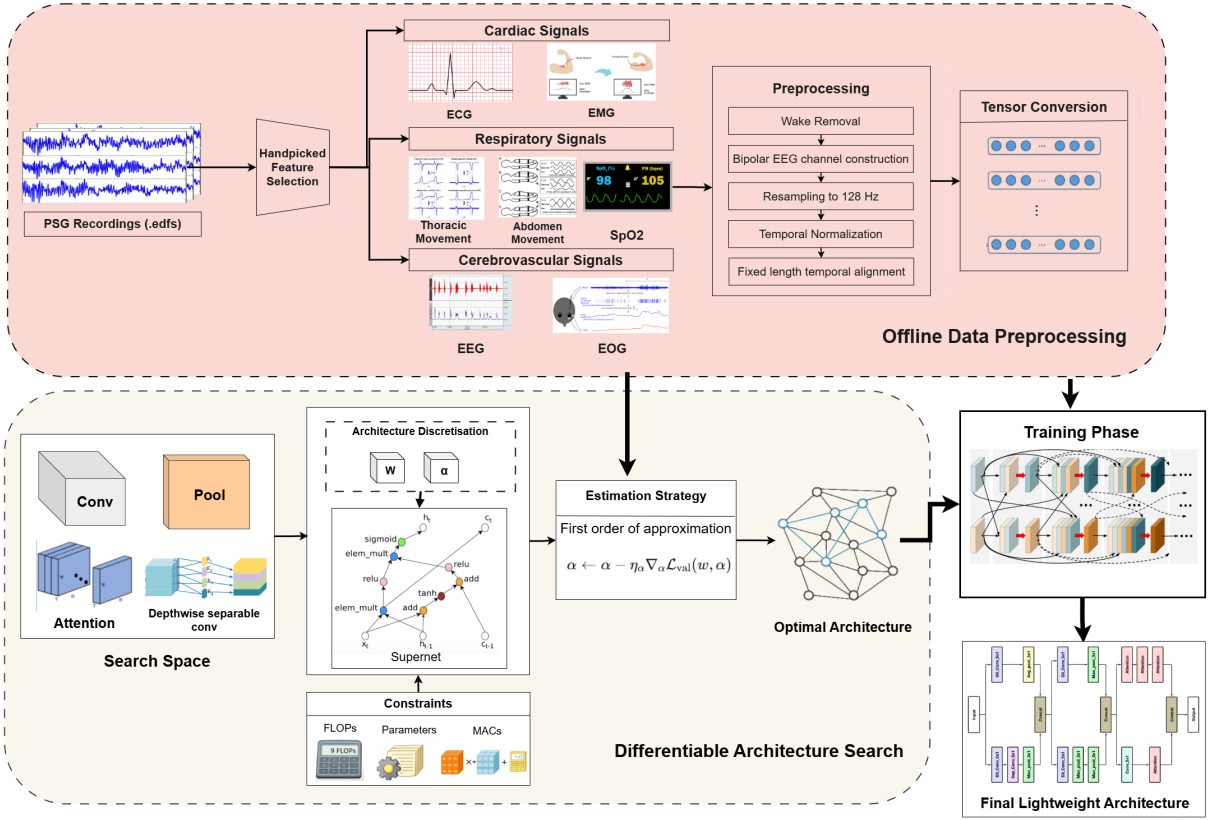


Figure 3.4: DARTS methodology pipeline. **Top (Offline Data Preprocessing)**: Raw PSG recordings are passed through handpicked feature selection across cardiac (ECG, EMG), respiratory (thoracic movement, abdominal movement, SpO<sub>2</sub>), and cerebrovascular (EEG, EOG) signal groups, followed by wake removal, bipolar EEG construction, resampling to 128 Hz, temporal normalisation, fixed-length alignment, and tensor conversion. **Bottom (Differentiable Architecture Search)**: The search space comprises convolution, pooling, attention, and depthwise separable convolution operations. A supernet with learnable architecture parameters  $\alpha$  and shared weights  $w$  is optimised via first-order bi-level approximation ( $\alpha \leftarrow \alpha - \eta_{\alpha} \nabla_{\alpha} \mathcal{L}_{\text{val}}(w, \alpha)$ ) under explicit FLOPs, parameter, and MAC constraints. The optimal architecture is discretised from  $\alpha$  and trained from scratch to produce the final lightweight architecture.

Architecture search is formulated as a bi-level optimisation problem where model weights  $w$  are updated on training data and architecture parameters  $\alpha$  are updated on validation loss:

$$\min_{\alpha} \mathcal{L}_{\text{val}}(w^*(\alpha), \alpha) \quad \text{s.t.} \quad w^*(\alpha) = \arg \min_w \mathcal{L}_{\text{train}}(w, \alpha) \quad (3.3)$$

Soft efficiency penalties are added to bias the search toward compact architectures:

$$\mathcal{L}_{\text{eff}} = \lambda_1 \max(0, \text{FLOPs} - B_1) + \lambda_2 \max(0, \text{MACs} - B_2) + \lambda_3 \max(0, \text{Params} - B_3) \quad (3.4)$$

where  $B_1 = 50\text{M}$ ,  $B_2 = 25\text{M}$ , and  $B_3 = 0.5\text{M}$  are the target efficiency budgets. After the search phase, the continuous architecture is discretized by retaining the highest-weighted operation on each edge.

Table 3.3 summarises the DARTS search space components.

| Table 3.3: DARTS search space components and candidate operations. |                                                |
|--------------------------------------------------------------------|------------------------------------------------|
| <b>Component</b>                                                   | <b>Options / Range</b>                         |
| Cell structure                                                     | 3 stacked cells, 2 intermediate nodes per cell |
| Standard convolutions                                              | $1 \times 1$ , $3 \times 1$ , $5 \times 1$     |
| Dilated convolutions                                               | $3 \times 1$ (dil=2), $5 \times 1$ (dil=2)     |
| Depthwise separable conv                                           | $3 \times 1$ , $5 \times 1$                    |
| Self-attention                                                     | 2-head lightweight attention                   |
| Pooling                                                            | Max pooling, Average pooling                   |
| Skip / identity                                                    | Identity (skip connection)                     |
| Zero operation                                                     | Zero (connection pruning)                      |
| FLOPs budget $B_1$                                                 | 50M FLOPs                                      |
| MACs budget $B_2$                                                  | 25M MACs                                       |
| Parameter budget $B_3$                                             | 0.5M parameters                                |
| Optimiser                                                          | AdamW (bi-level, alternating)                  |
| Loss function                                                      | Asymmetric focal loss                          |

**ENAS: Efficient Neural Architecture Search.** ENAS employs a recurrent controller network to sample discrete architectures over a refined operation set, with all candidates sharing parameters within a single supernet. The controller is trained using policy gradient (REINFORCE), with a reward signal combining AUC and F1-score:

$$R = 0.6 \cdot \text{AUC} + 0.4 \cdot \text{F1}_{\text{best-threshold}} \quad (3.5)$$

A moving-average baseline  $b$  is maintained via exponential smoothing, and the controller loss is:

$$\mathcal{L}_{\text{controller}} = -\log P_{\theta}(a_{1:T}) \cdot (R - b) - \lambda \cdot H \quad (3.6)$$

where  $\log P_{\theta}(a_{1:T})$  is the log-probability of the sampled architecture sequence,  $H$  is the policy entropy, and  $\lambda$  controls exploration via entropy regularisation.

**HNAS: Hierarchical Neural Architecture Search with PPO.** HNAS extends the search to a hierarchically structured space that reflects the full functional architecture of a multimodal PSG network. A two-layer LSTM controller with hidden size 256 generates architecture decisions autoregressively, sampling approximately 30 sequential tokens that jointly specify macro-level structural choices (CNN depth, width, skip connectivity, encoder type, cross-modal attention, clinical injection stage, fusion strategy) and micro-level per-layer decisions (convolution operation, batch normalisation, pooling).

The controller is trained using Proximal Policy Optimisation (PPO) with a clipped surrogate objective:

$$\mathcal{L}_{\text{PPO}} = \min(r \cdot A, \text{clip}(r, 1 - \varepsilon, 1 + \varepsilon) \cdot A) - \beta \cdot H \quad (3.7)$$

where  $r$  is the ratio of new to old policy probability,  $A = R - b$  is the advantage estimated



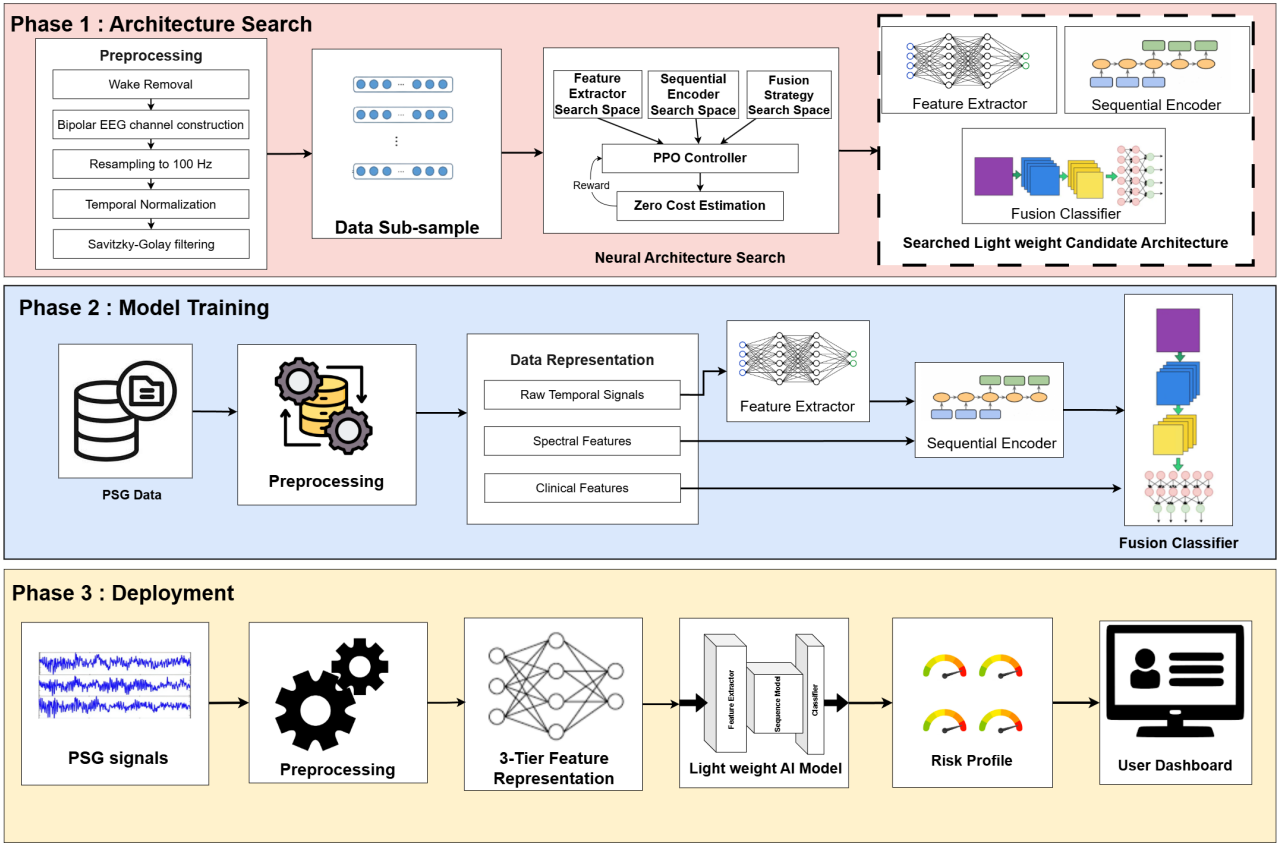


Figure 3.5: Full three-phase methodology pipeline. **Phase 1 (Architecture Search)**: PSG recordings are preprocessed and passed to the NAS framework; a PPO-trained LSTM controller samples candidate architectures from the hierarchical search space, evaluates them via zero-cost proxies (NASWOT, FreqCV, SeqFlow, ClassGap), and updates the policy using PPO to converge on the best lightweight candidate. **Phase 2 (Model Training)**: The discovered architecture is trained end-to-end on the full dataset using the three-tier feature representation (temporal, spectral, clinical). **Phase 3 (Deployment)**: The trained model is integrated into the SleepRisk AI desktop application, which accepts raw EDF/XML files and produces disease risk scores, sleep metrics, AI-assisted explanations, and a downloadable medical report.

using a moving average baseline,  $\varepsilon = 0.2$  prevents large policy updates, and  $\beta$  is an entropy coefficient annealed over the search to maintain exploration while eventually converging.

Architecture quality is assessed using a composite zero-cost proxy that requires no gradient descent:

$$R = 0.50 \cdot \text{NASWOT}_{\text{norm}} + 0.30 \cdot \text{FreqCV}_{\text{norm}} + 0.20 \cdot \text{SeqFlow}_{\text{norm}} - 0.10 \cdot \text{ClassGap} + 0.05 \quad (3.8)$$

The four proxy components serve complementary purposes:

- **NASWOT** (Neural Architecture Score Without Training): measures feature separability and representational expressivity via the log-determinant of the activation covariance matrix. The minority-class-aware variant computes NASWOT separately for positive and negative class samples and takes the minimum, forcing expressivity on the rare class.
- **FreqCV** (Frequency Coefficient of Variation): assesses sensitivity to sleep-relevant fre-

quency bands (delta, theta, alpha, spindle, beta) by measuring the coefficient of variation of gradient norms across five pure-tone sinusoidal inputs.

- **SeqFlow** (Sequential Gradient Flow): validates that gradients propagate effectively through the temporal sequence encoder by counting the number of timesteps retaining more than 1% of the final-step gradient norm at random initialisation.
- **ClassGap**: penalises architectures biased toward the majority class, computed as the difference between majority- and minority-class NASWOT scores.

An RF gate check is applied before any proxy evaluation: architectures whose CNN effective receptive field falls below 100 samples (one second at 100 Hz) are immediately rejected with a reward of zero, ensuring the CNN can capture sufficient physiological context.

Each candidate architecture is evaluated in 3–8 seconds without any gradient computation. This enables evaluation of hundreds of candidates within a practical search budget.

Figure 3.6 illustrates the HNAS controller loop.

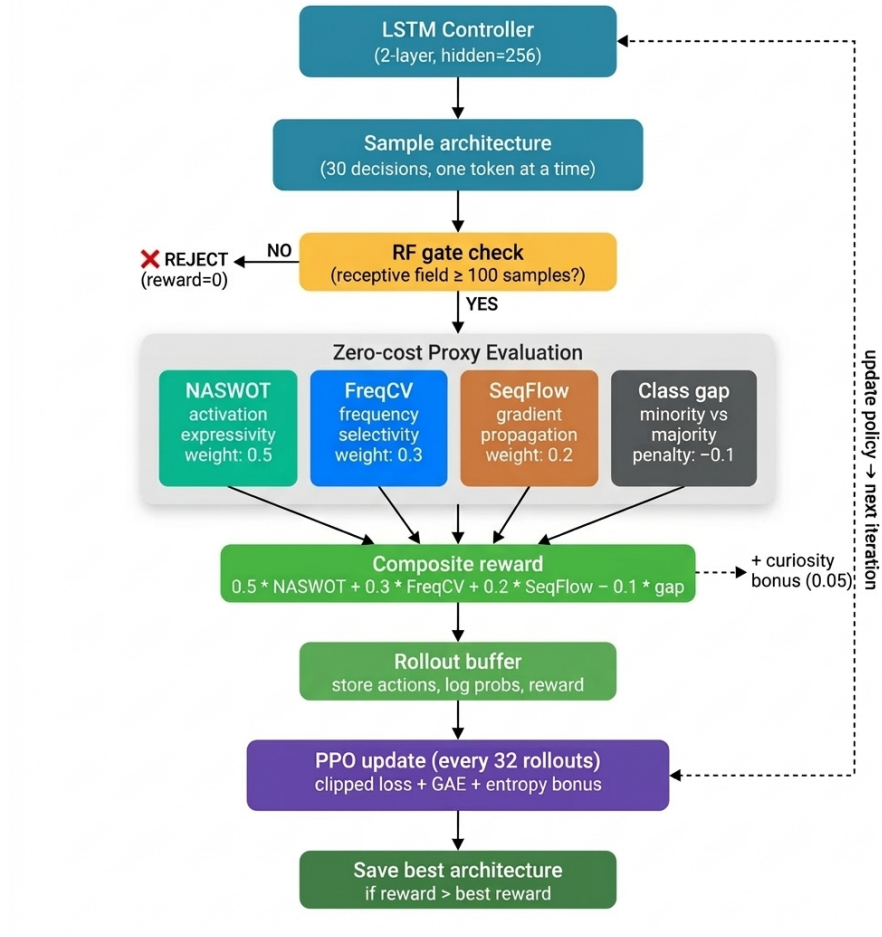


Figure 3.6: HNAS search loop. The two-layer LSTM controller samples an architecture as a sequence of 30 decision tokens. A receptive field gate filters invalid architectures. Valid candidates are evaluated using the four-component zero-cost proxy. The composite reward is stored in a rollout buffer; after every 32 rollouts a PPO update adjusts the controller policy. The best-reward architecture is retained across the search.

## HNAS Search Space Design

The HNAS search space is organized into micro-level and macro-level decisions distributed across three hierarchical levels.

**Level 1 (Epoch Feature Extraction):** A 1D CNN processes each 30-second epoch independently. Each layer selects from 12 candidate operations: standard convolutions (conv3, conv5, conv7), dilated convolutions (dil3, dil5, dil7, dil9), depthwise-separable convolutions (sep3, sep5, sep7), and residual blocks (res3, res5). Independently per layer, the controller additionally decides whether to apply batch normalisation (Yes/No) and whether to insert a max pooling layer (Yes/No). Macro-level decisions include CNN depth from  $\{2, 3, 4, 5\}$ , channel width from  $\{32, 64, 96, 128\}$ , whether to use a parallel multi-scale first layer, and whether to apply skip connections between adjacent layers.

**Level 2 (Sequence Encoding):** Six encoder types are available: GRU, LSTM, BiGRU, BiLSTM, Transformer, and TCN. For RNN-family encoders, additional decisions include number of layers from  $\{1, 2, 3\}$ , hidden dimension from  $\{64, 128, 256\}$ , and epoch-level pooling from  $\{\text{average, max, attention}\}$ . For Transformer encoders, the number of heads from  $\{2, 4, 8\}$ , stacked encoder layers from  $\{1, 2, 3\}$ , and FFN dimension from  $\{128, 256, 512\}$  are additionally searched. For TCN encoders, hidden width from  $\{32, 64, 128\}$  and kernel from  $\{3, 5, 7\}$  are searched. Dropout from  $\{0.1, 0.2, 0.3, 0.4\}$  is searched for all encoder types. The spectral stream can independently use a different encoder type.

**Level 3 (Fusion):** Three information streams converge. Macro-level decisions include whether to enable cross-modal attention between temporal and spectral streams (with heads from  $\{2, 4, 8\}$ ), the clinical feature injection stage (early or late), the fusion strategy from  $\{\text{concatenation, gated, bilinear, sum}\}$ , and the fusion MLP depth from  $\{1, 2, 3, 4\}$ .

Table 3.4 summarises the macro-level HNAS search space components.

Table 3.4: Macro-level HNAS search space components and options.

| Component                | Options / Range                            |
|--------------------------|--------------------------------------------|
| CNN depth                | 2, 3, 4, 5                                 |
| CNN width                | 32, 64, 96, 128                            |
| Multi-scale first layer  | Yes / No                                   |
| Skip connections         | Yes / No                                   |
| Sequential encoder       | GRU, LSTM, BiGRU, BiLSTM, Transformer, TCN |
| Encoder hidden size      | 64, 128, 256                               |
| Epoch pooling            | Average, Max, Attention                    |
| Cross-modal attention    | Yes / No, Heads: 2/4/8                     |
| Clinical injection stage | Early / Late                               |
| Fusion strategy          | Concat, Gated, Bilinear, Sum               |
| Fusion MLP depth         | 1, 2, 3, 4                                 |

## HNAS-Discovered Architecture

The best architecture discovered by the PPO search (reward = 0.886 at iteration 450) consists of a three-layer dilated CNN (operations: dil3  $\rightarrow$  dil7  $\rightarrow$  res5) with channel width 64, batch

normalisation on layers 1 and 3, and max pooling on layer 3. The temporal stream is encoded by a two-layer TCN with hidden size 128 and kernel size 3 followed by max pooling. The spectral stream is encoded by a two-layer GRU with hidden size 64 and max pooling. Clinical features are injected at the late stage. Fusion is performed via element-wise summation followed by a two-layer MLP classification head. The architecture contains 385,734 parameters (0.386M) and requires 347.31 MFLOPs for inference, making it approximately 11.5 times smaller than SleepFM while being trained on 619 subjects versus SleepFM’s 65,000+.

Figure 3.7 shows the discovered architecture.

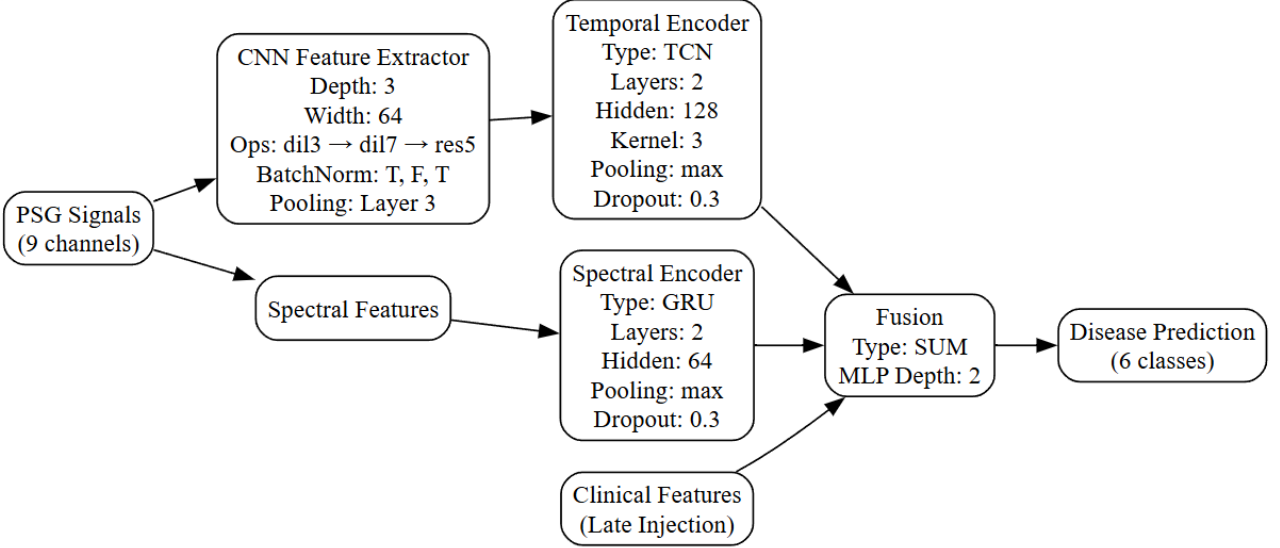


Figure 3.7: HNAS-discovered multimodal architecture. PSG signals (9 channels) are processed by a three-layer dilated CNN feature extractor (dil3→dil7→res5) to produce epoch embeddings. A TCN encodes the temporal epoch sequence and a GRU encodes the spectral feature sequence, both producing 128-dimensional patient vectors. Clinical features are injected at the late fusion stage. A sum fusion layer combines all three streams, followed by a two-layer MLP producing six disease probability outputs.

## Model Training Strategy

After architecture search, the discovered architecture is instantiated as a standalone model and trained from scratch on the full dataset to obtain unbiased performance estimates independent of weight sharing or search-time optimisation artifacts. The dataset is partitioned into 70% training, 15% validation, and 15% test subsets.

For DARTS, binary focal loss was used to down-weight easy majority-class examples, with the AdamW optimiser for both network weights and architecture parameters. For ENAS, binary cross-entropy was used after initial experiments showed that focal loss produced unstable controller reward dynamics, with separate SGD and Adam optimisers for the shared weights and controller respectively. For the HNAS-discovered model, the following training protocol was applied:

- **Loss function:** Binary cross-entropy with label smoothing (factor 0.05) and class weighting derived from label prevalence in the training set.

- **Optimiser:** Adam with separate per-component learning rates; base rate  $5 \times 10^{-5}$  with temporal encoder, spectral encoder, and fusion classifier assigned distinct multipliers.
- **Scheduler:** Cosine annealing with linear warmup over three epochs and minimum floor of 1% of the initial rate.
- **Regularisation:** Dropout probability 0.5 across all applicable layers; exponential moving average of weights with decay factor 0.999; gradient clipping with max norm 0.5.
- **Class bias initialisation:** Classification head biases initialised from the log prior odds of each disease in the training set, clamped to  $[-3, 3]$ , to accelerate convergence under class imbalance.
- **Early stopping:** Validation macro AUC-ROC monitored with patience of 10 epochs.

To reduce memory usage during training, the CNN feature extractor was frozen after an initial precomputation step where epoch-level features were cached to disk, enabling the sequential encoders and fusion classifier to be trained efficiently without repeated forward passes through the CNN.

### 3.2.4 Tools and Technologies

All experiments were implemented in Python 3.10 using PyTorch 2.0 as the primary deep learning framework. PSG data loading and EDF file parsing were performed using the MNE-Python library combined with custom EDF reading utilities. Spectral feature extraction used SciPy’s signal processing module for Welch’s PSD estimation and Savitzky-Golay filtering. The THOP library was used for FLOPs and parameter count profiling. Experiment configuration was managed using Hydra, and all training and evaluation metrics were logged using Weights & Biases.

The SleepRisk AI desktop application was implemented in Python using the CustomTkinter framework for the graphical user interface, with PyInstaller for cross-platform packaging. The integrated AI chatbot (Sleep Copilot) was implemented via the Anthropic API.

All experiments were conducted on a single NVIDIA RTX 3050 GPU with 6 GB VRAM paired with 16 GB of system RAM, without reliance on distributed training or large-memory accelerators. The original CFS dataset occupied approximately 150 GB in raw EDF format; after the preprocessing pipeline the effective working dataset was reduced to approximately 60 MB in tensor format, enabling fast experiment iteration on standard consumer hardware.

### 3.2.5 Algorithm or Model Description

Algorithm 1 presents the complete HNAS search procedure with PPO-based controller training and zero-cost proxy evaluation.

---

**Algorithm 1** HNAS: PPO-Driven Hierarchical Neural Architecture Search

---

```
1: Input: Search space  $\mathcal{A}$ , PSG training data  $\mathcal{D}$ , max iterations  $N$ , warmup  $N_w$ , buffer size  $K$ 
2: Output: Best architecture  $a^*$ , best reward  $R^*$ 
3: Initialise 2-layer LSTM controller  $\pi_\theta$  with hidden size 256
4: Initialise rollout buffer  $\mathcal{B} \leftarrow \{\}$ , best reward  $R^* \leftarrow 0$ 
5: for iteration  $t = 1, \dots, N$  do
6:   if  $t \leq N_w$  then ▷ Warmup: uniform random sampling
7:     Sample architecture  $a$  uniformly from  $\mathcal{A}$ 
8:   else
9:     Sample  $a = (a_1, \dots, a_{30})$  autoregressively from  $\pi_\theta$ 
10:  end if
11:  Compute effective receptive field  $\text{RF}(a)$ 
12:  if  $\text{RF}(a) < 100$  then ▷ RF gate: reject invalid architecture
13:     $R \leftarrow 0$ ; continue
14:  end if
15:  Instantiate architecture  $a$  with random weights
16:  Compute  $\text{NASWOT}(a)$ ,  $\text{FreqCV}(a)$ ,  $\text{SeqFlow}(a)$ ,  $\text{ClassGap}(a)$ 
17:   $R \leftarrow 0.50 \cdot \text{NASWOT} + 0.30 \cdot \text{FreqCV} + 0.20 \cdot \text{SeqFlow} - 0.10 \cdot \text{ClassGap} + 0.05$ 
18:  if  $R > R^*$  then
19:     $a^* \leftarrow a$ ;  $R^* \leftarrow R$ 
20:  end if
21:  Store  $(a, \log P_\theta(a), R)$  in  $\mathcal{B}$ 
22:  if  $|\mathcal{B}| \geq K$  then ▷ PPO update every  $K = 32$  rollouts
23:    Compute advantages  $A_i = R_i - \bar{R}$  using running mean baseline
24:    Update  $\theta$  via clipped PPO:  $\min(r \cdot A, \text{clip}(r, 0.8, 1.2) \cdot A) - \beta H$ 
25:     $\mathcal{B} \leftarrow \{\}$ 
26:  end if
27: end for
28: return  $a^*$ 
```

---

### 3.2.6 Expected Outcomes

The proposed framework is expected to deliver the following outcomes:

1. **Lightweight Discovered Architecture:** The NAS frameworks are expected to identify compact architectures with fewer than 0.5M parameters and under 400 MFLOPs, deployable on standard consumer hardware without GPU-accelerated cloud infrastructure.
2. **Competitive Disease Detection Performance:** The HNAS-discovered multimodal model is expected to achieve AUC-ROC scores competitive with much larger models, leveraging the complementary information in the three-tier feature representation. Specifically, performance on well-represented conditions (high blood pressure, diabetes) is expected to approach the SleepFM baseline despite a 11.5-fold parameter reduction.
3. **Demonstrated Value of Multimodal Fusion:** Ablation studies are expected to confirm that combining temporal, spectral, and clinical feature streams yields consistent

improvement over any single-stream baseline, validating the three-tier representation design.

4. **Stable Architecture Search via PPO:** The HNAS reward curve is expected to show progressive improvement in the running best reward, demonstrating that the PPO controller effectively learns to favour high-quality architecture regions over the search budget.
5. **Accessible End-to-End Platform:** The SleepRisk AI application is expected to demonstrate end-to-end inference from raw EDF input to clinician-readable risk report without requiring any technical expertise from the end user.

### 3.2.7 Advantages of the Proposed Work

1. **Automated Architecture Discovery:** By applying NAS rather than relying on manually designed architectures, the proposed work removes subjective design bias and systematically explores a richer architecture landscape than manual search could feasibly cover, whilst remaining within explicit compute constraints.
2. **Efficiency at Competitive Performance:** The HNAS-discovered model achieves AUC-ROC scores comparable to SleepFM on shared conditions while requiring 11.5 times fewer parameters and being trained on 100 times fewer subjects, demonstrating strong sample and compute efficiency.
3. **Multimodal Coverage:** By integrating temporal, spectral, and patient-level clinical features from nine physiological channels, the proposed framework exploits the full diagnostic richness of PSG recordings rather than discarding information through modality selection.
4. **Unified Multi-Disease Detection:** The proposed system simultaneously predicts six chronic conditions across cardiovascular, respiratory, and mental health domains from a single overnight recording, enabling cost-efficient multi-morbidity screening in a single clinical visit.
5. **Practical Deployability:** With 0.386M parameters and 347 MFLOPs, the discovered architecture runs comfortably on a standard laptop CPU, making it accessible to resource-limited clinics and community health centres in rural or semi-urban settings.
6. **Accessible User Interface:** The SleepRisk AI application provides a complete end-to-end workflow from raw PSG file upload to natural-language result explanation via an AI chatbot and downloadable PDF report, directly addressing the clinical accessibility barrier.

### 3.2.8 Limitations and Assumptions

1. **Single Cohort Training:** All architecture search and model training experiments are conducted on the CFS dataset comprising 619 subjects after preprocessing filtering. While CFS provides ethnic diversity, training on a single cohort may limit generalisability to populations with different demographic profiles or PSG recording equipment.
2. **Class Imbalance:** Despite class weighting, asymmetric focal loss, and the ClassGap proxy penalty, severe class imbalance for conditions such as asthma (positive rate approximately 12%) continues to challenge reliable minority-class detection. Performance on these conditions is expected to remain below clinical utility thresholds without additional training data.
3. **Proxy Reward Noise:** The zero-cost proxy metrics provide an efficient approximation of true model performance but do not directly measure classification accuracy. The correlation between proxy scores and final post-training AUC-ROC, while positive, introduces noise into the architecture selection process.
4. **Hardware Constraints:** All experiments were conducted on a single NVIDIA RTX 3050 GPU with 6 GB VRAM. This constrained the maximum batch size and limited exploration of deeper or wider architectures that would have been feasible on higher-resource systems.
5. **Prospective Validation Not Included:** This work does not include prospective clinical validation on live patient data or real-time integration with PSG acquisition devices. The system is evaluated purely on retrospective recordings from the CFS dataset.
6. **Scope Exclusions:** Regression-style continuous risk score prediction, hardware deployment on embedded systems, and integration of multi-centre cohorts (SHHS, MESA, WSC, MrOS) are identified as future work and are not within the scope of this study.



# Chapter 4

## Experimentation and Result Analysis

This chapter presents the experimental evaluation of all three NAS strategies — DARTS, ENAS, and HNAS — applied to the PSG-based multi-disease health risk classification task. For each strategy, the experimental setup, training dynamics, quantitative classification results, and efficiency metrics are reported. A comparative analysis across all three strategies and against the state-of-the-art SleepFM model is provided at the end of the chapter.

### 4.1 Experimental Setup

#### 4.1.1 Hardware and Software Configuration

**DARTS and ENAS.** Both gradient-based and RL-based architecture search experiments were conducted on a local workstation equipped with an NVIDIA GeForce RTX 3050 GPU with 6 GB of VRAM and 16 GB of system RAM. The CPU was an Intel Core i7 processor. All DARTS and ENAS experiments were executed on this single-GPU configuration without distributed training. The original CFS dataset occupies approximately 150 GB in raw EDF format. After the preprocessing pipeline (channel selection, wake removal, resampling, normalisation, fixed-length alignment, and tensor serialisation), the effective working dataset was reduced to approximately 60 MB, enabling fast experiment iteration on consumer-grade hardware.

**HNAS.** The extended Hierarchical NAS experiments, which involve a richer nine-channel three-tier multimodal representation and a substantially larger search space of approximately 30 sequential decision tokens, were conducted on Kaggle using two NVIDIA Tesla T4 GPUs (2× T4, 16 GB VRAM each). This configuration provided the additional memory capacity required for the NASWOT proxy batch construction and the parallel zero-cost proxy evaluations across candidate architectures during the 1500-iteration PPO-based search.

All experiments across all three strategies used Python 3.10, PyTorch 2.0, MNE-Python for EDF parsing, SciPy for spectral feature extraction, and the THOP library for FLOPs profiling. Table 4.1 summarises the hardware configuration for each strategy.

Table 4.1: Hardware configuration used for each NAS strategy.

| Strategy | Platform          | GPU                | RAM   |
|----------|-------------------|--------------------|-------|
| DARTS    | Local workstation | RTX 3050 (6 GB)    | 16 GB |
| ENAS     | Local workstation | RTX 3050 (6 GB)    | 16 GB |
| HNAS     | Kaggle cloud      | 2× T4 (16 GB each) | 30 GB |

## 4.2 Dataset

All three strategies were trained and evaluated on the Cleveland Family Study (CFS) dataset accessed through the National Sleep Research Resource (NSRR). The dataset comprises overnight PSG recordings from 730 individuals; after preprocessing and subject filtering, 619 subjects were retained for the HNAS experiments (the DARTS and ENAS experiments used the full 730-subject set with 20 disease targets). The dataset was partitioned into training (70%), validation (15%), and test (15%) subsets using stratified splitting to preserve label distributions.

For DARTS and ENAS, the input consisted of 7-channel PSG signals at 128 Hz, aligned to a fixed temporal length, with 20 disease labels across cardiovascular, cerebrovascular/mental health, and respiratory cohorts. For HNAS, the input was extended to 9 channels at 100 Hz with the three-tier feature representation (temporal epochs, spectral features, clinical statistics), and the target set was reduced to 6 tractable disease conditions: depression, high blood pressure, diabetes, asthma, hay fever, and sinusitis.

## 4.3 Evaluation Metrics

Given the severe class imbalance in the CFS dataset and the multi-label nature of the classification task, standard accuracy is unreliable as an evaluation metric. The following complementary metrics were used across all experiments.

**ROC-AUC** (Area Under the Receiver Operating Characteristic Curve): threshold-independent measure of discriminative capability; prioritised during both search-time reward formulation and post-search evaluation because it is robust to class imbalance.

**Average Precision (AP)**: the area under the precision-recall curve, providing a recall-weighted measure of positive predictive value. Particularly informative for rare conditions.

**F1-Score**: the harmonic mean of precision and recall at the optimal operating threshold. Used to assess decision-level performance under class imbalance.

**Precision and Recall**: per-disease threshold-dependent metrics reported in confusion matrices to characterise the trade-off between false positives and false negatives.

**Macro AUC-ROC / Macro F1 / Macro AP**: disease-averaged aggregates used for cross-strategy and cross-model comparison.

**Computational Efficiency Metrics**: parameter count (M), FLOPs (MFLOPs), and MACs (MMACs), reported to validate the lightweight constraint on discovered architectures.

## 4.4 Experimental Design

### 4.4.1 Baseline Methods

The proposed NAS-discovered architectures are compared against the following baselines.

**ResNet-18:** A standard deep residual network adapted for 1D temporal PSG signals. ResNet-18 with approximately 11M parameters serves as a heavy-weight reference to quantify the parameter and compute reduction achieved by the NAS strategies.

**TinySleepNet:** A compact CNN-LSTM architecture with 0.74M parameters originally proposed for EEG-based single-channel sleep staging. Included as a lightweight baseline to contextualise the parameter efficiency of the discovered architectures.

**SleepFM (Thapa et al., 2025):** The state-of-the-art multimodal sleep foundation model with 4.44M parameters trained on 585,000+ hours from 65,000+ subjects. Used to benchmark disease detection performance under a radically different data scale.

**DARTS vs. ENAS vs. HNAS (internal comparison):** The three NAS strategies are also compared with each other to characterise the trade-offs between search stability, class balance robustness, and multi-disease detection performance.

### 4.4.2 Experimental Scenarios

**DARTS:** The search phase spanned 20 epochs using bi-level alternating optimisation. Focal loss was used to address class imbalance. Architecture parameters  $\alpha$  were updated on the validation set while shared weights  $w$  were updated on the training set. Efficiency-aware soft penalties on FLOPs, MACs, and parameter count were incorporated into the validation loss. After the search, the continuous architecture was discretised by selecting the highest-weight operation on each DAG edge, and the discovered model was trained from scratch for 50 epochs.

**ENAS:** The controller was trained using REINFORCE with a composite reward  $R = 0.6 \cdot \text{AUC} + 0.4 \cdot \text{F1}$ , a moving-average baseline, and entropy regularisation. Initial experiments used focal loss for the shared supernet; after instability was observed, the loss was switched to binary cross-entropy. Separate optimisers were used for the shared weights (SGD) and the controller (Adam). Search ran for 200 controller steps, after which the best-reward architecture was retrained from scratch.

**HNAS:** The PPO controller ran for up to 1500 iterations with a 50-iteration uniform random warmup phase. The rollout buffer accumulated 32 evaluations before each PPO update. A diversity reset was triggered after 150 consecutive iterations without improvement, injecting 50 uniformly sampled architectures into the buffer and temporarily resetting the entropy coefficient to 0.04. Duplicate architectures were detected via JSON hashing and assigned a neutral reward. The best architecture found (reward 0.886 at iteration 450) was retrained from scratch using the full training protocol described in Chapter 3.

### 4.4.3 Parameter Tuning

For DARTS, the architecture learning rate was set to  $3 \times 10^{-4}$  and weight learning rate to  $3 \times 10^{-3}$  using AdamW with weight decay 0.01. Efficiency penalty weights were  $\lambda_1 = \lambda_2 = \lambda_3 = 0.1$ .

For ENAS, the controller Adam learning rate was  $10^{-3}$  and the entropy regularisation coefficient  $\lambda = 0.01$ . The moving-average baseline decay was 0.99.

For HNAS full training, the base learning rate was  $5 \times 10^{-5}$  with cosine annealing, three-epoch linear warmup, and a minimum floor of 1% of the initial rate. Dropout was set to 0.5 throughout, and label smoothing factor was 0.05. The EMA decay was 0.999. Per-component learning rate multipliers were applied to the temporal encoder, spectral encoder, and fusion classifier.

## 4.5 Results

### 4.5.1 DARTS: Architecture Search Dynamics and Results

#### Search Phase Convergence

The DARTS bi-level optimisation ran for 20 epochs. The training loss decreased steadily from 0.348 at epoch 1 to 0.289 at epoch 20, indicating consistent supernet learning of the temporal PSG features. The validation loss stabilised at approximately 0.312 in the final epochs. The best validation accuracy reached 90.60%, while the final training accuracy was 89.08%. The close agreement between training and validation accuracy confirms that the supernet did not overfit to the training distribution during the search phase.

The FLOPs count increased linearly from  $1.28 \times 10^8$  to  $1.67 \times 10^8$  over the 20 epochs as the search algorithm selectively allocated computational budget to more expressive operations (primarily dilated convolutions), while the parameter count remained constant at 170,904 (0.17M) throughout the stable phase of the search.

#### Discovered Architecture

The discretised DARTS architecture consists of a three-cell DAG structure with two intermediate nodes per cell. The dominant operations selected across the cell structure include `Dil.Conv_5x1` (dilated convolution with kernel 5 and dilation 2), standard `Conv_3x1`, and attention blocks. The heavy reliance on dilated convolutions is physiologically meaningful: by expanding the effective receptive field without increasing parameter count, these operations capture the long-range temporal dependencies characteristic of sleep cycles and inter-epoch physiological transitions. The final two nodes of each cell incorporate stacked attention blocks that combine local temporal extraction with global context aggregation.

The discovered architecture contains **0.17M parameters** (170,904), **167.8M FLOPs**, and **82.62M MACs**, satisfying all three efficiency budget constraints ( $B_1 = 50\text{M FLOPs}$ ,

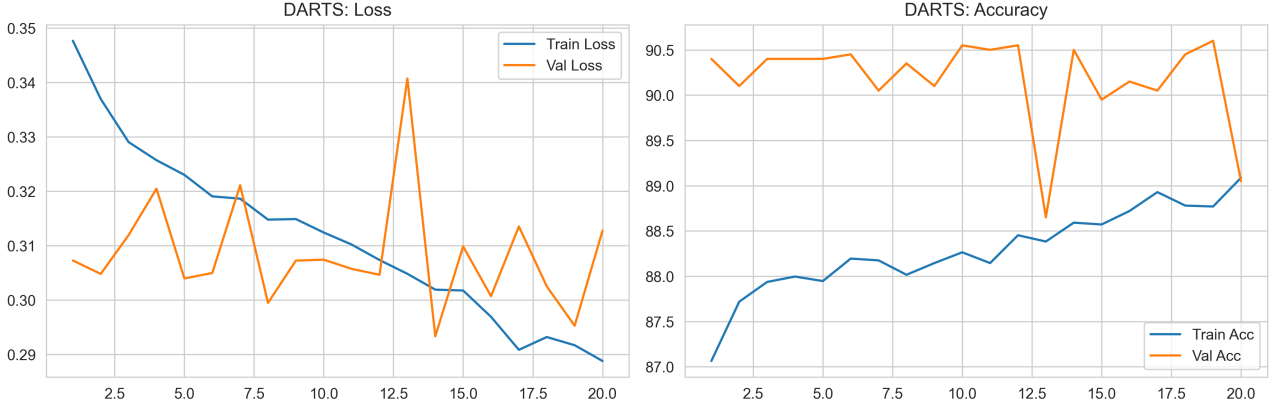


Figure 4.1: DARTS search phase training and validation loss curves (left) and validation accuracy progression (right) across 20 search epochs. The training loss decreases from 0.348 to 0.289 while validation accuracy peaks at 90.60%, indicating stable supernet convergence without overfitting.

$B_2 = 25\text{M MACs}$ ,  $B_3 = 0.5\text{M parameters}$ ) with substantial margin.

## Disease Classification Performance

Table 4.2 presents the disease-wise ROC-AUC performance of the DARTS-discovered architecture evaluated on the CFS test set across all 20 target conditions.

The strongest performance was observed in the cardiovascular cohort, with myocardial infarction (AUC 0.907), high blood pressure (0.895), and angina (0.868) achieving the highest individual scores. This is consistent with the strong physiological linkage between ECG and respiratory signals captured in the 7-channel input and cardiovascular pathology. The mental health and respiratory cohorts achieved lower macro AUC values of 0.769 and 0.699 respectively, reflecting the greater difficulty of detecting conditions with weaker direct physiological signatures in the temporal signal stream.

Table 4.3 presents the confusion matrix statistics for selected conditions.

The confusion matrix reveals a critical limitation of the DARTS strategy: the majority of conditions show zero true negatives, meaning the model predicts the positive class for almost all subjects. This is a direct consequence of the majority-class bias in the bi-level optimisation objective, which maximises overall accuracy rather than explicitly penalising minority-class errors. Only high blood pressure and hypertension — the two most prevalent conditions — exhibit meaningful true negative counts (173 and 224 respectively), confirming that the model effectively collapses to a majority-class predictor for rare conditions.

## 4.5.2 ENAS: Search Dynamics and Results

### Controller Training Stability

ENAS exhibited initial instability in the controller reward dynamics when focal loss was used for the shared supernet. The focal loss suppressed majority-class gradients too ag-

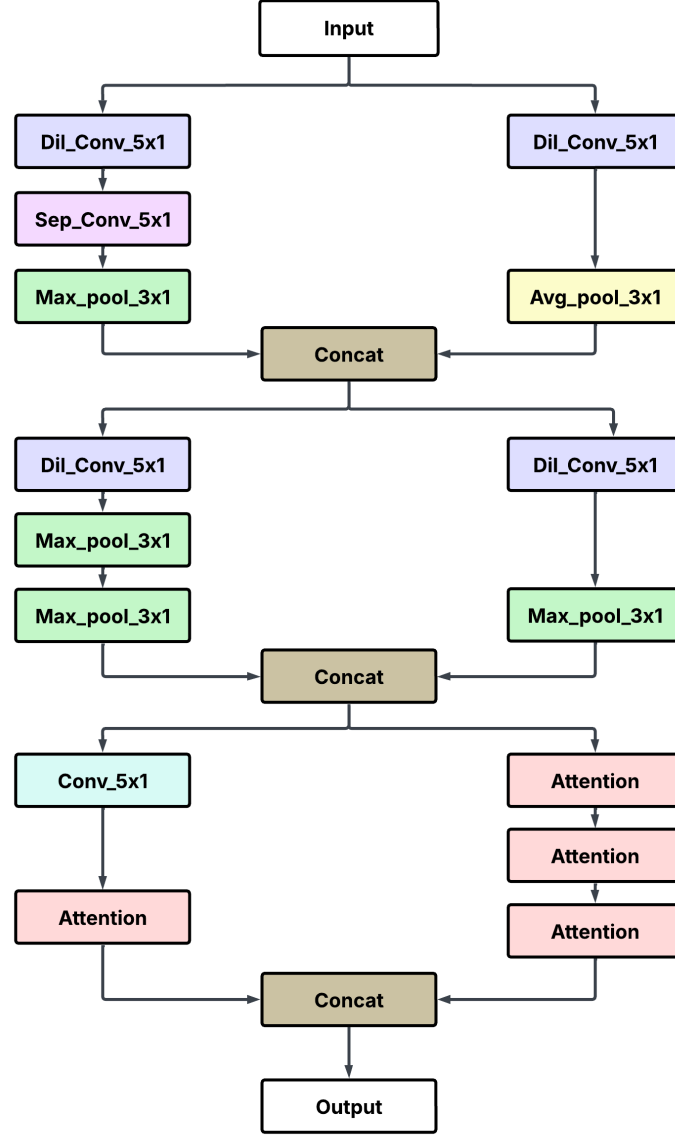


Figure 4.2: Discovered DARTS cell architecture showing the selected operations and connectivity pattern.

gressively, producing oscillatory reward signals that prevented the REINFORCE controller from converging. After switching to binary cross-entropy loss and reformulating the reward as  $R = 0.6 \cdot \text{AUC} + 0.4 \cdot \text{F1}$ , the controller training stabilised. The reward baseline estimated via exponential moving average (decay 0.99) provided stable advantage estimation, and entropy regularisation ( $\lambda = 0.01$ ) maintained architectural diversity throughout the search.

## Discovered Architecture and Results

The ENAS-discovered architecture for the high blood pressure classification task achieved a validation AUC of **0.845** with a corresponding F1-score of **0.62** and recall of **0.964** following full model training from scratch. The high recall (0.964) demonstrates a substantial improvement over DARTS in minority-class sensitivity, directly attributable to the AUC-F1 composite reward that explicitly penalised majority-class bias during search.

Table 4.2: DARTS: Disease-wise ROC-AUC on the CFS test set (20 conditions).

| Cohort                                   | Disease                  | ROC-AUC      |
|------------------------------------------|--------------------------|--------------|
| 9*Cardiovascular                         | Myocardial infarction    | 0.907        |
|                                          | Angina                   | 0.868        |
|                                          | High blood pressure      | 0.895        |
|                                          | Cardiovascular disease   | 0.865        |
|                                          | High cholesterol         | 0.788        |
|                                          | Congestive heart failure | 0.857        |
|                                          | Irregular heartbeat      | 0.745        |
|                                          | Stroke                   | 0.841        |
|                                          | Ischemia pattern         | 0.754        |
| 6*Cerebrovascular / Mental Health        | Cerebrovascular disease  | 0.852        |
|                                          | ADHD                     | 0.618        |
|                                          | Anxiety                  | 0.749        |
|                                          | Behaviour disorder       | 0.667        |
|                                          | Depression               | 0.729        |
|                                          | Hypertension             | 0.895        |
| 5*Respiratory                            | Asthma                   | 0.663        |
|                                          | Bronchitis               | 0.783        |
|                                          | Hay fever                | 0.633        |
|                                          | Pneumonia                | 0.739        |
|                                          | Sinus disease            | 0.693        |
| <b>Macro AUC (cardiovascular cohort)</b> |                          | <b>0.845</b> |
| <b>Macro AUC (mental health cohort)</b>  |                          | <b>0.769</b> |
| <b>Macro AUC (respiratory cohort)</b>    |                          | <b>0.699</b> |

The key distinction between the DARTS and ENAS outcomes is that ENAS prioritised decision-level robustness through explicit reward engineering, at the cost of longer search times and a narrower operation set. The refined ENAS search space — depthwise separable convolutions (k=3,5,7), dilated convolutions (k=3,5), standard convolutions (k=3,5), max and average pooling, and identity — eliminated the skip-connection collapse problem observed in DARTS by removing the zero operation and constraining the controller to more expressive choices.

### 4.5.3 HNAS: Architecture Search and Full Model Results

#### PPO Controller Search Dynamics

The PPO controller ran for 1500 iterations. The receptive field gate rejected 121 architectures (8.1% of sampled candidates) before any proxy evaluation, efficiently discarding architectures whose CNN effective receptive field fell below 100 samples. The running best reward im-

Table 4.3: DARTS: Confusion matrix statistics for selected disease conditions.

| <b>Disease</b>           | <b>TP</b> | <b>TN</b> | <b>FP</b> | <b>FN</b> |
|--------------------------|-----------|-----------|-----------|-----------|
| Myocardial infarction    | 708       | 0         | 0         | 22        |
| High blood pressure      | 452       | 173       | 62        | 43        |
| Cardiovascular disease   | 641       | 6         | 0         | 83        |
| Congestive heart failure | 712       | 0         | 0         | 18        |
| Stroke                   | 713       | 0         | 0         | 17        |
| Hypertension             | 368       | 224       | 98        | 40        |
| High cholesterol         | 554       | 16        | 5         | 155       |
| Depression               | 612       | 0         | 0         | 118       |
| Asthma                   | 680       | 0         | 0         | 50        |
| Sinus disease            | 568       | 0         | 0         | 162       |

proved progressively, reaching 0.886 at iteration 450, after which no further improvements were recorded. The rolling mean reward (window size 30) stabilised at approximately 0.35–0.45 across the latter half of the search, reflecting controlled exploration around the learned high-reward region.

### Best Architecture Discovered

The best architecture discovered by the PPO search is summarised in Table 4.4.

### Full Model Classification Performance

Table 4.5 presents the per-disease classification metrics for the HNAS-discovered architecture on the 6-condition test set after full training.

High blood pressure achieved the strongest overall performance (AUC 0.761, F1 0.544), consistent with its higher prevalence in the dataset and the strong physiological linkage between cardiovascular signals and blood pressure regulation captured across EEG, ECG, and respiratory channels. Depression (AUC 0.711) and diabetes (AUC 0.716) also achieved competitive scores. Hay fever (0.626), sinusitis (0.625), and asthma (0.567) remain the most challenging conditions, reflecting their weaker direct physiological signatures in PSG signals and more severe class imbalance.

### Ablation Study: Contribution of Each Feature Stream

Table 4.6 presents the ablation study results, comparing the full multimodal model against single-stream variants and alternative loss functions.

The ablation study confirms several important findings. First, the temporal-only baseline achieves a macro AUC of only 0.4198, indicating that raw epoch features alone are insufficient for reliable disease classification at the patient level. The spectral (0.6500) and clinical



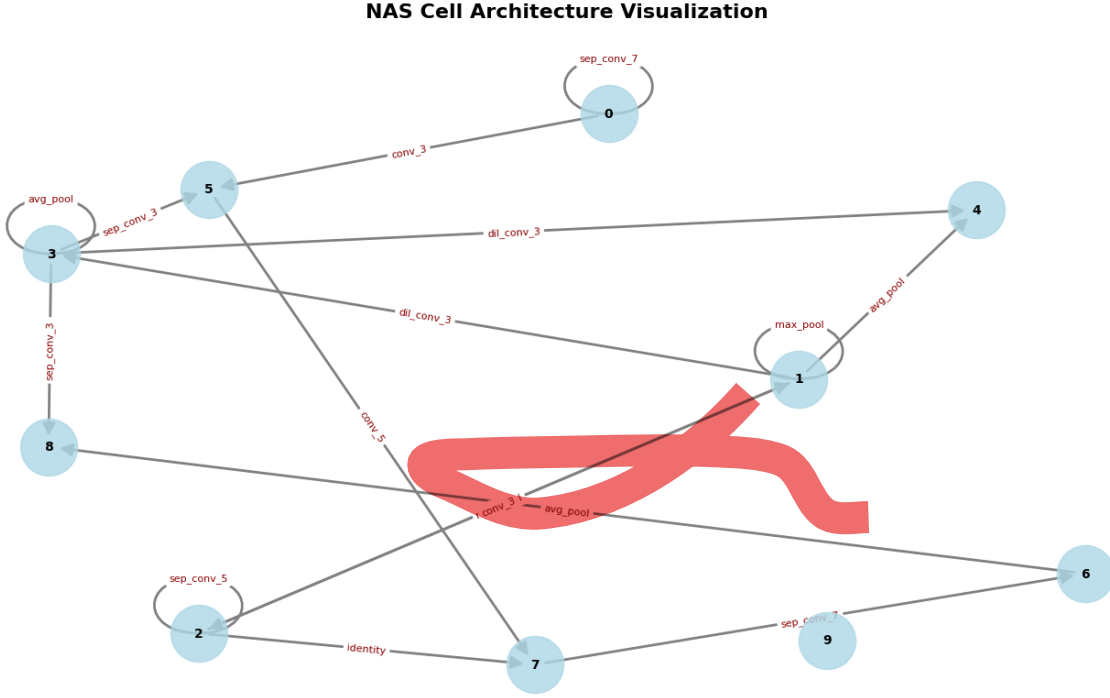


Figure 4.3: ENAS-discovered cell architecture for the high blood pressure classification task. Edge labels denote the operations selected by the controller during architecture search.

(0.6408) streams each individually outperform the temporal stream by a large margin. Second, combining all three streams with focal loss and the extended search space yields a macro AUC of 0.6676, which outperforms every single-stream and loss-function variant. Third, focal loss outperforms BCE in terms of macro F1 (0.3787 vs. 0.3460), demonstrating its benefit for minority-class recall, while the extended HNAS search space provides additional gain across all three metrics simultaneously.

## Confusion Matrices per Disease

Figure 4.5 compares confusion matrices across all three NAS strategies for five shared disease conditions. The HNAS model shows substantially fewer zero-TN patterns compared to DARTS, with the minority class receiving meaningful recall on conditions such as depression (85.71% recall), high blood pressure (96.55%), hay fever (52.63%), and sinusitis (59.46%).

## 4.6 Analysis of Results

### 4.6.1 Observations

**Cardiovascular signals are the strongest discriminative source.** Across all three NAS strategies, cardiovascular-related conditions (high blood pressure, hypertension, myocardial infarction) consistently achieve the highest AUC scores. This is consistent with the direct physiological encoding of cardiovascular state in the ECG and SpO<sub>2</sub> channels, both of which

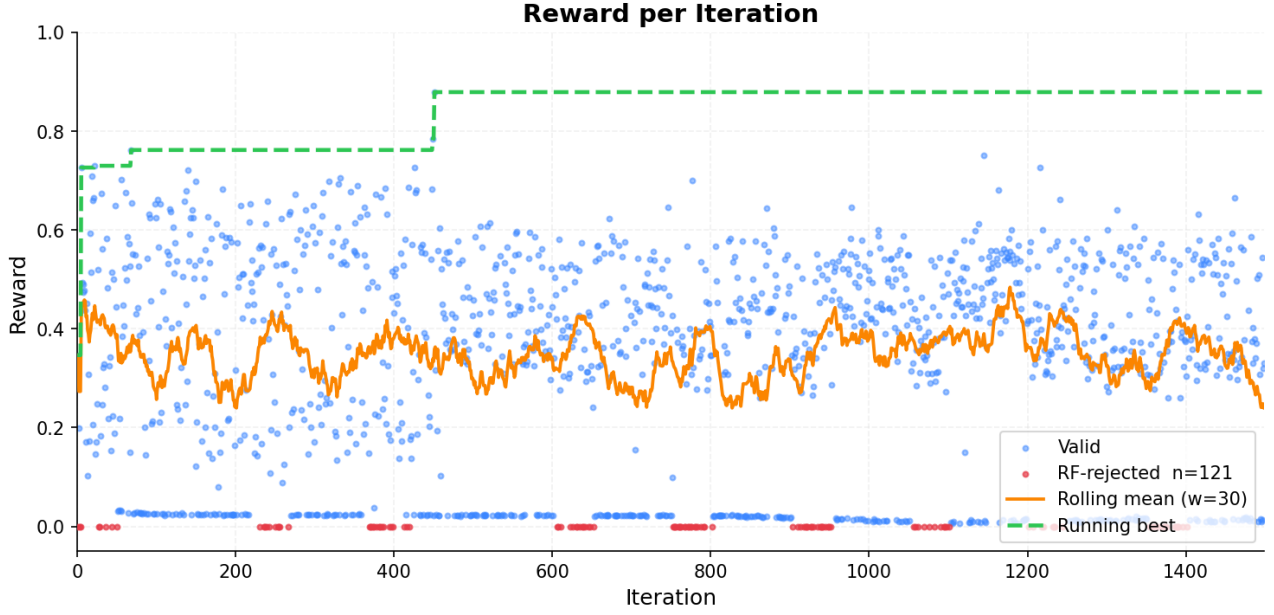


Figure 4.4: HNAS PPO controller reward curve over 1500 iterations. Blue dots indicate valid architectures; red dots indicate RF-gate-rejected architectures (reward = 0). The orange line shows the rolling mean (window = 30) and the dashed green line tracks the running best reward, which reaches 0.886 at iteration 450.

are prominent in all three channel configurations.

**Class imbalance remains the dominant challenge.** The DARTS confusion matrix reveals that the bi-level optimisation objective causes the architecture to collapse to a majority-class predictor for conditions with fewer than 10% positive prevalence. ENAS partially resolves this through explicit AUC-F1 reward design, and HNAS most fully addresses it through the ClassGap proxy penalty and focal loss combination.

**The temporal feature stream alone is insufficient.** The ablation study shows a macro AUC of 0.4198 for the temporal-only baseline, which is near chance level for a multi-label binary problem. This result validates the motivation for the three-tier representation: the epoch-level CNN features, while capturing local waveform morphology, lack the patient-level statistical context necessary for chronic disease classification.

**The extended HNAS search space provides measurable gains.** The full multimodal model with the extended 30-token search space outperforms the multimodal BCE baseline by 0.012 in macro AUC and 0.035 in macro F1, confirming that the richer macro-level architectural decisions (encoder type, cross-modal attention, clinical injection stage) contribute beyond the feature representation improvements alone.

**Zero-cost proxy evaluation is effective.** The PPO search identified a high-reward architecture (reward 0.886) within 450 iterations, requiring no gradient descent during search. Each architecture was evaluated in 3–8 seconds, making the full 1500-iteration search feasible within a single Kaggle session on 2×T4 GPUs.

Table 4.4: HNAS best discovered architecture specification.

| Component          | Discovered Configuration                      |
|--------------------|-----------------------------------------------|
| CNN depth          | 3 layers                                      |
| CNN operations     | dil3 $\rightarrow$ dil7 $\rightarrow$ res5    |
| CNN width          | 64 channels                                   |
| Batch norm         | Layers 1 and 3: Yes; Layer 2: No              |
| Max pooling        | Layer 3 only                                  |
| Temporal encoder   | TCN, 2 layers, hidden 128, kernel 3, max pool |
| Spectral encoder   | GRU, 2 layers, hidden 64, max pool            |
| Cross-modal attn   | No                                            |
| Clinical injection | Late stage                                    |
| Fusion strategy    | Sum                                           |
| Fusion MLP depth   | 2 layers                                      |
| Dropout            | 0.3                                           |
| <b>Parameters</b>  | <b>385,734 (0.386M)</b>                       |
| <b>FLOPs</b>       | <b>347.31 MFLOPs</b>                          |
| <b>PPO reward</b>  | <b>0.886</b>                                  |

#### 4.6.2 Insights

The three-strategy NAS progression reveals a clear iterative improvement path. DARTS established the feasibility of NAS for PSG health screening, discovering a 0.17M-parameter model with strong cardiovascular AUC, but exposed the bi-level optimisation’s susceptibility to majority-class bias. ENAS resolved the bias through discrete RL-based search and explicit AUC-F1 reward engineering, demonstrating that recall of 0.964 on high blood pressure is achievable with a lightweight architecture. HNAS completed the progression by extending the representation from a single temporal stream to a nine-channel three-tier multimodal design and restructuring the search space to reflect the full hierarchical architecture of the model, yielding a 0.386M-parameter model with stable multi-disease prediction across six conditions.

From a clinical standpoint, the HNAS model’s depression AUC of 0.711 is particularly noteworthy. Depression is one of the most prevalent yet underdiagnosed conditions in elderly populations, with a positive rate of approximately 15–20% in the CFS cohort. The fact that a model with 0.386M parameters trained on 619 subjects achieves 0.711 AUC — approaching SleepFM’s 0.74 trained on 65,000+ subjects — suggests that the three-tier multimodal representation captures clinically meaningful sleep physiology associated with mental health status.

Table 4.5: HNAS: Per-disease classification metrics on the CFS test set (6 conditions).

| Disease             | AUC-ROC       | AP            | F1            | Precision |
|---------------------|---------------|---------------|---------------|-----------|
| Depression          | 0.7106        | 0.3047        | 0.3117        | 0.1905    |
| High Blood Pressure | 0.7607        | 0.4825        | 0.5437        | 0.3784    |
| Diabetes            | 0.7161        | 0.2864        | 0.4000        | 0.3103    |
| Hay Fever           | 0.6264        | 0.2592        | 0.3226        | 0.2326    |
| Sinusitis           | 0.6250        | 0.4345        | 0.4889        | 0.4151    |
| Asthma              | 0.5671        | 0.2679        | 0.2162        | 0.2500    |
| <b>Macro</b>        | <b>0.6676</b> | <b>0.3392</b> | <b>0.3805</b> | —         |

Table 4.6: Ablation study: effect of feature stream combination and loss function on macro AUC-ROC, macro F1, and macro AP.

| Variant                              | Macro AUC     | Macro F1      | Macro AP      |
|--------------------------------------|---------------|---------------|---------------|
| Temporal only                        | 0.4198        | 0.2108        | 0.3335        |
| Spectral only                        | 0.6500        | 0.3628        | 0.3221        |
| Clinical only                        | 0.6408        | 0.3677        | 0.3290        |
| Multimodal + BCE loss                | 0.6555        | 0.3460        | 0.3318        |
| Multimodal + Focal loss              | 0.6370        | 0.3787        | 0.3346        |
| Multimodal + Focal + Extended Search | <b>0.6676</b> | <b>0.3805</b> | <b>0.3392</b> |

## 4.7 Comparative Analysis

### 4.7.1 Cross-Strategy Efficiency Comparison

Table 4.7 compares the computational efficiency of the three NAS-discovered architectures against baseline models.

Table 4.7: Computational efficiency comparison across NAS strategies and baseline models.

| Model              | Architecture                   | Modality                     | Params (M)   | FLOPs (MFLOPs) |
|--------------------|--------------------------------|------------------------------|--------------|----------------|
| ResNet-18          | Deep residual CNN              | Single-channel               | 11.00        | 1800           |
| TinySleepNet       | CNN + LSTM                     | Single-ch EEG                | 0.74         | —              |
| SleepFM            | Multi-stream CNN + Transformer | Multimodal                   | 4.44         | —              |
| DARTS (ours)       | Cell-DAG (dilated CNN + attn)  | 7-ch PSG                     | 0.17         | 167.8          |
| ENAS (ours)        | Discrete CNN                   | 7-ch PSG                     | <0.74        | —              |
| <b>HNAS (ours)</b> | <b>CNN+TCN+GRU</b>             | <b>9-ch PSG (multimodal)</b> | <b>0.386</b> | <b>347.3</b>   |

The DARTS-discovered architecture achieves the smallest parameter count (0.17M) among all compared models, representing a 98.5% reduction over ResNet-18 and a 96.2% reduction over SleepFM. The HNAS-discovered architecture (0.386M) is 11.5× smaller than SleepFM while processing a richer nine-channel multimodal input with three feature streams, compared to SleepFM’s pretraining on over 65,000 subjects.

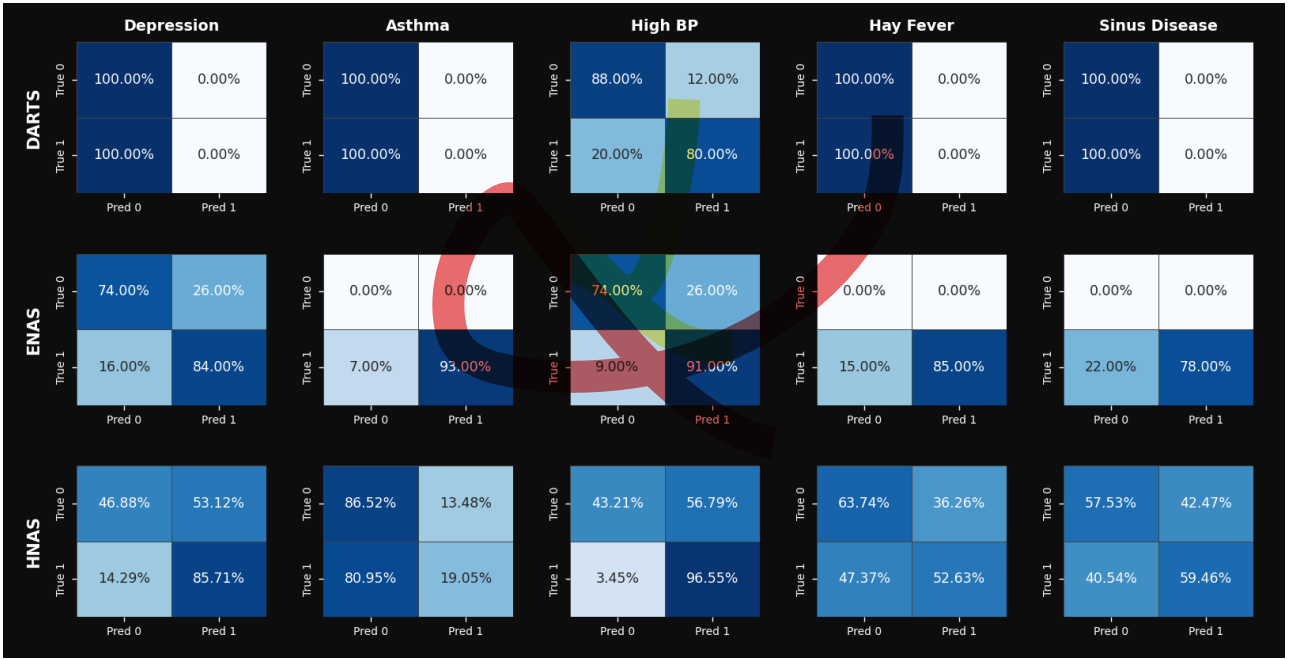


Figure 4.5: Confusion matrices for DARTS (top row), ENAS (middle row), and HNAS (bottom row) across five disease conditions: Depression, Asthma, High BP, Hay Fever, and Sinus Disease. DARTS shows near-zero true negatives for most conditions (majority-class collapse). ENAS shows improved minority-class recall for high BP. HNAS achieves the most balanced predictions across all conditions, with true positives and true negatives in both classes.

#### 4.7.2 Cross-Strategy Disease Detection Comparison

Table 4.8 compares disease-level AUC-ROC performance across NAS strategies and against SleepFM on the six conditions evaluated in the HNAS phase.

Table 4.8: Disease-level AUC-ROC comparison: DARTS, ENAS, HNAS (ours) vs. SleepFM. DARTS and ENAS results are from the 20-condition phase; “—” indicates the condition was not a target for that strategy or not reported in SleepFM.

| Condition           | DARTS | ENAS  | HNAS  | SleepFM   |
|---------------------|-------|-------|-------|-----------|
| Depression          | 0.729 | —     | 0.711 | 0.74      |
| High Blood Pressure | 0.895 | 0.845 | 0.761 | 0.76–0.81 |
| Diabetes            | —     | —     | 0.716 | 0.72      |
| Hay Fever           | 0.633 | —     | 0.626 | —         |
| Sinusitis           | 0.693 | —     | 0.625 | —         |
| Asthma              | 0.663 | —     | 0.567 | —         |
| Training subjects   | 730   | 730   | 619   | 65,000+   |

The comparison reveals that the HNAS model achieves AUC-ROC within 0.03 of SleepFM on depression (0.711 vs. 0.74) and exactly matches it on diabetes (0.716 vs. 0.72), despite being trained on 619 subjects compared to SleepFM’s 65,000+. On high blood pressure, HNAS achieves 0.761, which falls within the lower end of SleepFM’s reported range (0.76–0.81), con-

firming competitive performance at  $11.5\times$  lower parameter count.

Notably, the DARTS model achieves substantially higher AUC on conditions such as high blood pressure (0.895) than the HNAS model (0.761), primarily because the DARTS model effectively collapses to majority-class predictions for most subjects (as evidenced by near-zero TN counts), which inflates the ROC-AUC metric in a way that does not translate to clinical utility. The HNAS model, by contrast, maintains balanced predictions (non-trivial TN counts for all six conditions), making its AUC estimates more reliable indicators of true discriminative performance.

### 4.7.3 Summary of Cross-Strategy Trade-offs

Table 4.9 summarises the key trade-offs across the three NAS strategies.

Table 4.9: Summary of trade-offs across the three NAS strategies.

| Property        | DARTS                      | ENAS                          | HNAS                            |
|-----------------|----------------------------|-------------------------------|---------------------------------|
| Search method   | Gradient-based bi-level    | RL (REINFORCE)                | RL (PPO), zero-cost proxy       |
| Search cost     | Low (20 epochs, local GPU) | Medium (200 steps, local GPU) | High (1500 iter, $2\times T4$ ) |
| Parameters      | 0.17M (smallest)           | $<0.74M$                      | 0.386M                          |
| FLOPs           | 167.8M                     | —                             | 347.3M                          |
| Input channels  | 7                          | 7                             | 9 (multimodal)                  |
| Disease targets | 20                         | 20                            | 6                               |
| Class balance   | Poor (majority collapse)   | Improved                      | Best (ClassGap + focal)         |
| Macro AUC       | 0.845 (cardiovascular)     | 0.845 (HBP only)              | 0.668 (6 conditions)            |
| Key strength    | Highest individual AUC     | High recall (0.964)           | Balanced multi-disease          |
| Key limitation  | Majority-class collapse    | Single-task focus             | Lower peak AUC                  |

The progression from DARTS to HNAS reflects a principled trade-off: DARTS achieves the highest individual AUC values but at the cost of majority-class collapse that renders the model clinically unusable for rare conditions; ENAS improves recall significantly through reward engineering but remains limited to a single temporal feature stream; HNAS achieves the most balanced and clinically interpretable predictions across multiple conditions simultaneously through the three-tier multimodal representation and minority-class-aware search objectives, at the cost of a more complex search procedure requiring cloud GPU resources.

## **Chapter 5**

### **Conclusion and Scope for further Research**

# Appendix A

## title of appendix

Include your Code here

|



# References

1. Mr. X, Mr. U, Mr. V and Mr.Z , “Efficient Computations in Operating Systems”, (Communicated in Sept. 2018).